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Glomerular Cell Biology and Podocytopathies

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CHAPTER OUTLINE

KEY POINTS

- The glomerulus represents a functional and integrated syncytium of four types of glomerular cells, which together ascertain glomerular filtration.
- Together, podocytes and glomerular endothelial cells allow for a size and charge selective glomerular filtration due to their specialized three-dimensional structure, extensive glycocalyx coating, and the coordinate synthesis and compression of the unique glomerular basement membrane.
- Mesangial cells provide structural support of the glomerular capillaries, regulate glomerular filtration, maintain glomerular endothelial health, and clear the mesangial space through phagocytosis.
- Parietal epithelial cells build the Bowman capsule to prevent leakage of the primary urine filtrate to the tubulointerstitium, contribute to glomerular scarring, and are thought to constitute a potential reservoir for podocytes in development, maturation, and eventually in adulthood.
- Hallmarks of diseases related to primary podocyte injury of genetic or autoimmune origin are proteinuria (often nephrotic range), foot process effacement, podocyte hypertrophy, and depletion.
- Crosstalk between glomerular cell types attenuates and/or perpetuates glomerular injury.
- Most forms of glomerular injury result from humoral and cellular immunologic mechanisms to which the glomerulus reacts by basic responses such as cellular proliferation, changes in glomerular cell phenotypes, and increased deposition of extracellular matrix.

Clinical Relevance

1. The glomerulus is a functional syncytium of four cell types, which interact in physiologic and also pathologic situations, thereby perpetuating and extending the site of glomerular injury.
2. Proteinuria is the hallmark of glomerular injury and should initiate an extensive workup by a nephrologist with focus on genetic, inflammatory, toxic, tumor, and infectious causes.
3. The site of glomerular injury determines the clinical picture. Whereas involvement of podocytes presents as glomerular injury with proteinuria or nephrotic syndrome, involvement of glomerular endothelial cells, the glomerular basement membrane and/or mesangial cells typically presents with microhematuria or nephritic syndrome.

GLOMERULAR CELL ANATOMY AND INJURY RESPONSE PATTERNS

Loss of protein into the urine (proteinuria), especially albumin (albuminuria), is the hallmark of glomerular disease and an important prognostic marker for a wide variety of kidney diseases, such as the numerically and economically increasingly challenge of diabetic nephropathy.[1] Albuminuria is also an independent risk factor for cardiovascular mortality.[2] Therefore understanding the pathophysiology of albuminuria and therapeutic approaches to its modification has major clinical and health economic significance.

The kidney of a healthy 70-kg adult filters approximately 180 L of plasma per day. Filtration and urine production

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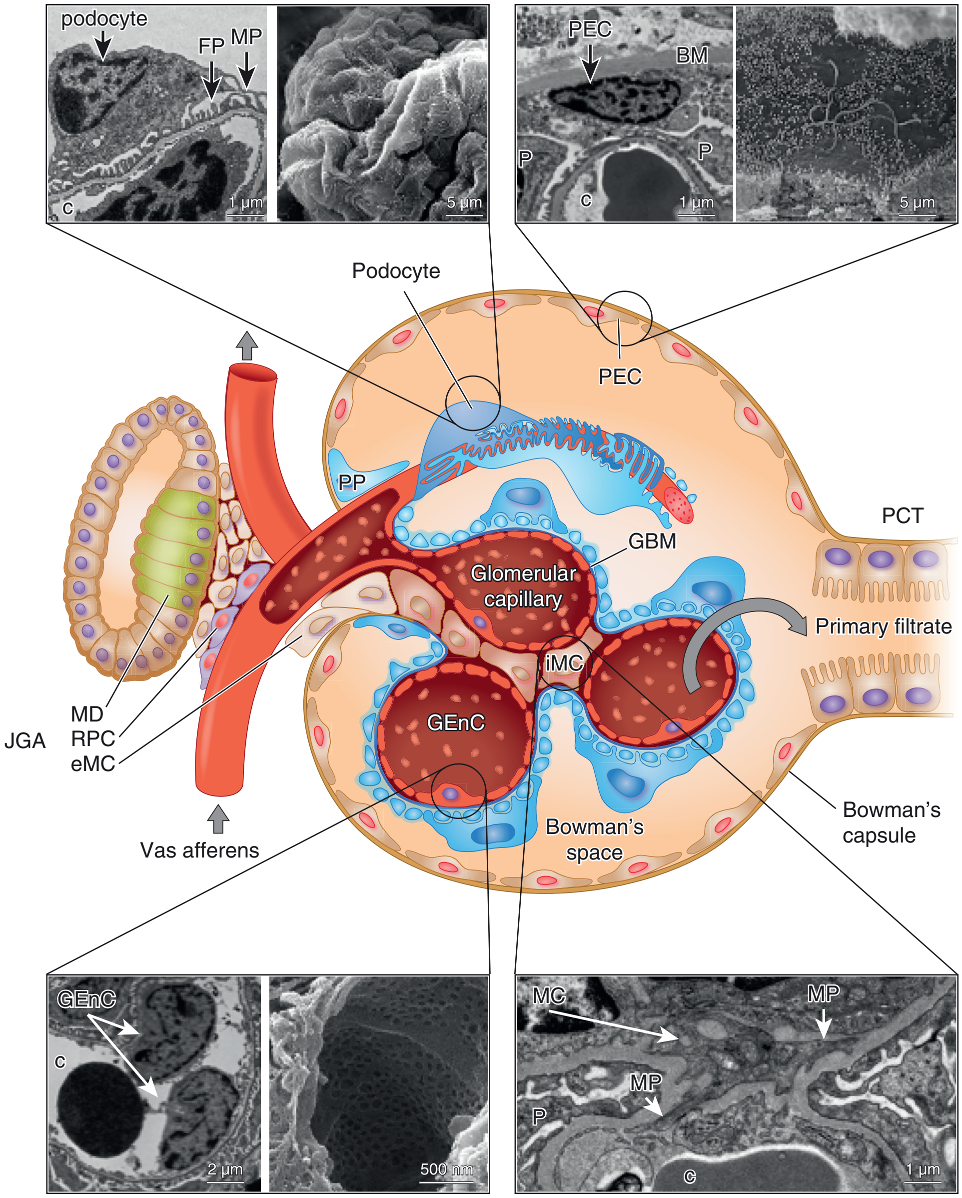

Fig. 4.1 Scheme of the glomerulus depicting the localization of the four resident glomerular cells, namely podocytes, parietal epithelial cells (PECs), glomerular endothelial cells (GEnCs), and mesangial cells (MCs). Transmission and scanning electron micrographs show the typical ultrastructure of the glomerular cells. *DCT,* distal convoluted tubule. *GBM,* glomerular basement membrane; *GEnC,* glomerular endothelial cell; *iMC,* intraglomerular mesangial cell; *JGA,* juxta glomerular apparatus; *MD,* macula densa; *RPC,* renin producing cell, *eMC,* extraglomerular mesangial cell; *PCT,* proximal convoluted tubule; *PEC,* parietal epithelial cell; *PP,* parietal podocyte. Abbreviations in transmission electron micrographs: *BM,* Bowman membrane; *c,* capillary lumen; *FP,* foot process; *MP,* major process; *P,* podocyte. (All TEMs and SEMs are from Oliver Kretz, Hamburg, Germany.)

take place in the smallest functional unit of the kidney, called the nephron. In healthy humans, the average number of nephrons is ~1 million (range 250,000 to <2.5 million). The ultimate site of filtration is located at the beginning of the nephron in the renal corpuscle. The renal corpuscle (Fig. 4.1) comprises the glomerulus, a network (tuft) of capillaries, which is surrounded by parietal epithelial cells (PECs) of the Bowman capsule (BC) and thereby separated from the tubular system. As described and depicted in detail in Chapter 3, blood enters the glomerulus at the vascular pole through an afferent arteriole of the renal circulation and is drained into an efferent arteriole. The unique resistance of these arterioles generates a high pressure within the capillaries of the glomerulus, ultimately driving ultrafiltration of a primary urinary filtrate through the glomerular filtration barrier into the Bowman space with an effective filtration pressure of approximately 50 mm Hg.[3] Tracer studies point to a charge selectivity favoring the filtration of positively charged solutes.[4,5] Primary urine exits the renal corpuscle at the urinary pole to be further processed to final urine in the downstream tubular system. While water and small molecules such as glucose, salt, and amino acids freely pass across the glomerular filtration barrier, a partial impermeability to large molecules such as albumin into the primary urine is maintained.

The degree of fractional albumin filtration has been a matter of debate, but more recent 2-photon in vivo imaging studies indicate values for a glomerular sieving coefficient

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of 0.02 to 0.04 for albumin filtration into the primary urine, which for the most part is reabsorbed by the proximal tubule.[6] This partial impermeability to large molecules is achieved by a highly complex interplay of two types of glomerular cells, the visceral epithelial cells (podocytes) and glomerular endothelial cells (GEnCs), which ultimately compose the three-layered glomerular filtration barrier (podocyte—the 250–400 nm thick glomerular basement membrane [GBM]—endothelial cell). The GBM is composed of several types of extracellular matrix macromolecules (laminin, type IV collagen, the heparan sulfate proteoglycan agrin, and nidogen), which produce an interwoven meshwork thought to impart both size-selective and charge-selective properties.[7] Finally, intraglomerular mesangial cells occupy the space between the glomerular filtration barrier to provide structural support. As specialized pericytes, mesangial cells indirectly participate in filtration by reducing the glomerular surface area by contraction and are also thought to participate in matrix turnover and innate immune function.[8] While all glomerular cells form a functional and integrated syncytium, we describe each component separately as follows.

PODOCYTES

STRUCTURE

Podocytes are highly differentiated, mesenchymal-derived cells.[9] An apicobasal polarity axis allows for podocyte orientation between the urinary space and GBM.[10] Podocytes reside in the urinary space and embrace the glomerular capillaries with their flat cell body, from which they extend long branching major processes. Major processes give rise to secondary processes (often called foot processes), which interdigitate in a zipper-like fashion with foot processes of neighboring podocytes[11] (Fig. 4.2). In general, interdigitating foot processes of the mature podocyte are connected by bicellular junctions of uniform widths; however, areas of tricellular junctions (connection between foot processes from three different podocytes) have been described.[12] Foot processes are not randomly organized on glomerular capillaries but show a preferred orientation with foot processes being aligned in parallel with the axis of orientation (longitudinal axis).[13] Ridgelike prominences, which are formed at the basal surface of the cell body and major processes, serve as an adhesion apparatus for the attachment of cell body and major processes to the GBM and as a connecting apparatus of peripheral foot processes to the cell body and major processes.[12] Foot processes attach podocytes to the underlying extracellular matrix (ECM) of the GBM by specific proteins such as adhesion receptors including integrins, syndecans, vinculin, talin, and dystroglycan.[14] Besides the adhesion complex proteins, podocyte antigens such as PLA_2R1[15] and THSD7A[16] and extracellular proteases such as ADAM10[17] are expressed at the base of foot processes.[18] Foot processes are interconnected by slit diaphragms, which represent highly sophisticated cell–cell contacts that form an adjustable, nonclogging barrier. While the term "diaphragm" implicates a thin-layered sieve, recent studies revealed that the slit diaphragm is composed of multiple layers of flexible transmembrane molecules to limit the passage of macromolecules.[19] Structurally, the slit diaphragm combines components of several types of cell–cell junctions including tight, adhesion, neuronal, and gap junctions. Proteins from tight (ZO-1, JAM4, occludin, and cingulin) and adherens junctions (P-cadherin, FAT1, and the catenin family of proteins) are associated with the immunoglobulin superfamily members nephrin and neph1.[20] Nephrin and neph1 form the core component of the slit diaphragm by a bipartite assembly with neph1 molecules spanning the lower part of the slit close to the GBM and nephrin molecules positioned in the apical side.[19] Application of cryoelectron tomography of vitreous lamellae from high-pressure frozen native murine glomeruli now suggests that the molecular bases of slit diaphragms are nephrin-neph1 heterodimers, which form a flexible fishnet pattern through a complex interaction pattern with multiple contact sites between the molecules.[21,21a] The stomatin protein family member podocin (NPHS2) helps anchor nephrin to the plasma membrane and generates a signaling hub in lipid-rich membrane compartments (e.g., the Ca^{2+}-permeable transient receptor potential channel [TRPC]-6,[21] which might translate mechanical tension to ion channel action and cytoskeletal regulation).[22] On their intracellular C-terminal parts, nephrin and neph1 are associated with several signaling adaptor molecules and scaffold proteins linking the slit diaphragm to the actin cytoskeleton.[19,23]

Combining our recent knowledge derived from improved microscopy techniques, such as cryoelectron tomography,[19] as well as from high-resolution proteomic analysis of slit diaphragms affinity isolated from rodent kidney,[24] it is thought that the slit-covering layer of nephrin/neph1 molecules most likely does not operate as a filtration barrier[19,25] but rather is endowed with context-dependent dynamics via its coassembled protein network. As such, in the native slit diaphragm, nephrin, neph1, and podocin coassemble with distinct classes of proteins including components with enzymatic activity (MERTK), receptor-triggered signaling (ANPRC), and scaffolding function (ITM2B and its interacting partner A423) rather than with cell–cell junctional proteins.[24] In general, the podocyte cytoskeleton is highly elaborate. The podocyte cell body and major and foot processes contain vimentin-rich intermediate filaments that assist in maintaining cell shape and rigidity. Large microtubules form organized structures along major processes. Foot processes contain long actin fiber bundles that run cortically and contiguously to link adjacent podocytes. Actin, α-actinin-4, and myosin form a contractile system in foot processes, regulated by the interplay of the actin-binding proteins synaptopodin[26] and α-actinin-4 with Rho GTPases. This well-orchestrated actin and microtubule cytoskeleton ensures a high plasticity of the podocyte process network.[27,28] The apical surfaces of podocytes are covered by the surface sialomucin podocalyxin,[29] which has highly negative charge functions to keep adjacent foot processes separated, thereby keeping the urinary filtration barrier open.[30]

Glucose is the main source of energy for podocytes,[31] which is taken up through glucose transporters (GLUTs) and metabolized through anaerobic glycolysis.[32] Podocyte homeostasis strongly depends on the endocytic[33] and secretory[34] pathways, as well as the proteasome degradation system.[35,36] In aging, podocyte homeostasis is assured by autophagy.[37]

FUNCTION

The main podocyte function is to build, maintain, and regulate the three-layered glomerular filtration barrier.

1. **The synthesis and remodeling of the mature GBM** requires the crosstalk of podocytes with endothelial

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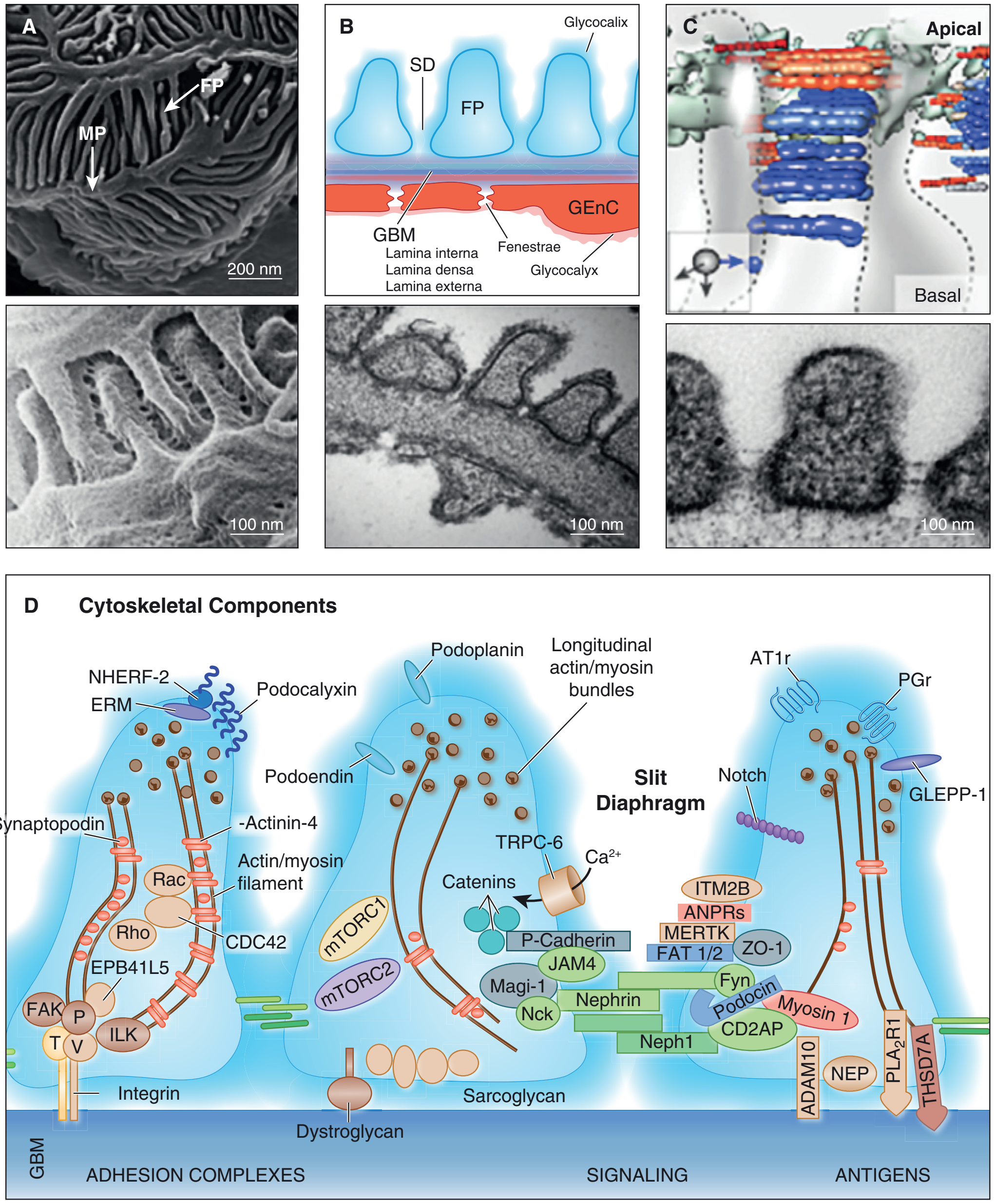

Fig. 4.2 Anatomy of the glomerular filter. (A) Scanning electron microscopy (SEM) of podocyte major processes (MPs) and secondary interdigitating foot processes (FPs). (B) Scheme and transmission electron microscopy (TEM) of three-layered glomerular filtration barrier consisting of 1. podocyte FP covered by a glycocalyx, the specialized cell–cell contact called slit diaphragm (SD) connecting the interdigitating foot processes; 2. the three-layered glomerular basement membrane consisting of podocyte-derived lamina interna, the podocyte/endothelial cell–derived lamina densa and the endothelial-derived lamina externa; and 3. the glomerular endothelial cells (GEnC) with the large fenestrae and thick glycocalyx. (C) Scheme of the typical sagittal view of two foot processes, showing three-layers of ~25 nm long strands, which are attributed to neph1 (blue color and 5 IG repeats) and one layer of ~40 nm long strands that represent nephrin (red color, and 9 IG repeats). (D) Scheme of podocyte foot process proteins constituting 1. foot process cytoskeleton (actin/myosin filaments, the actin binding proteins synaptopodin, α-actinin-4); 2. the podocyte adhesion complex (α3/ β1Integrin connecting the GBM to focal adhesions constituted of FAK = focal adhesion kinase, ILK = integrin linked kinase, T = talin, V = vinculin, P = paxillin, EPB41L5 = Ferm domain protein EPB41L5); 3. matrix interacting proteins dystroglycan, sarcoglycan; 4. the slit diaphragm (with the transmembrane proteins nephrin, neph1, FAT1/2 and P-cadherin, and the scaffolding protein ZO-1 = zonula-occludens 1, the intracellular signaling hub constituted of podocin, TRPC6 = transient receptor potential cation channel, subfamily C, member 6 and the connection to the actin cytoskeleton through CD2AP = CD2-associated protein, the cytoskeletal adaptor protein Nck, the multidomain scaffolding protein MAGI-1, and the junctional cell adhesion protein JAM4), and with signaling molecules such as receptor kinase MERTK, atrial natriuretic peptide receptors (ANPRs, receptor-mediated signaling), ITM2B = integral membrane protein 2B (amyloid precursor protein regulation); 5. the negatively charged sialoprotein podocalyxin (which is connected to the plasma membrane by NHERF-2 = Na+/H+ exchanger regulatory factor 2 and ERM = ezrin/moesin/radixin proteins) localizes to the surface of the plasma membrane, as do GLEPP-1 = glomerular epithelial cell membrane protein-tyrosine phosphatase 1, podoplanin, and podoendin; Of note, GLEPP-1 was recently also identified within the slit diaphragm protein network. 6. the G-coupled receptors AT1R = angiotensin receptor 1 and PGR = prostaglandin receptor; 7. the podocyte antigens, PLA_2R1 = phospholipase A2 receptor 1, THSD7A = thrombospondin type 1 domain-containing 7A , and NEP = neutral endopeptidase identified in antenatal and adult membranous nephropathy; 8. the extracellular protease ADAM10; and 9. intracellular signaling molecules Rho, Rac, mTORC1/2 = mammalian target of rapamycin complex 1 and 2, Notch and Fyn. (A and C: All TEM and SEM from mouse glomeruli are from Oliver Kretz, Hamburg, Germany. B: From Onions KL, Gamez M, Buckner NR, et. al. VEGFC reduces glomerular albumin permeability and protects against alterations in VEGF receptor expression in diabetic nephropathy. *Diabetes.* 2019; 68(1):172–187.)

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cells.[38] Podocytes and endothelial cells both secrete laminin 111 and type IV collagen α1 α2 α1 in GBM development and the final laminin 521 isoforms after maturation. However, only podocytes secrete type IV collagen α3 α4 α5 of the fully mature GBM.[39] Podocyte expression of prolyl 3-hydroxylase 2 (P3H2), which hydroxylates the 3' of prolines in type IV collagen subchains in the endoplasmic reticulum, is required for GBM remodeling. Mutations or loss of P3H2 protein results in thin basement membrane nephropathy.[40]

2. **Podocytes maintain the glomerular filtration barrier** by secreting survival factors such as angiopoietin-1 (binds to Tie2 on GEnC and MCs),[41] normosialylated angiopoietin-like-4 (binds to integrin αV β5 on GEnC)[42] and vascular endothelial growth factor A (binds to VEGFR2 on GEnC),[43] and stromal-derived factor 1 (binds to CXCR4 on GEnC),[44] which exert paracrine effects across the filtration barrier on glomerular endothelial and mesangial cells supporting their respective migration, differentiation, and survival.[45]
3. **Podocytes stabilize the glomerular filtration barrier** by expressing cell-matrix adhesion receptors such as integrin α3 β1, which connects laminin 521 in the GBM through various adaptor proteins to the intracellular actin cytoskeleton or integrins α2 β1 and αV β3, α-dystroglycan, syndecan-4, and type XVII collagen.[46]
4. **Podocytes regulate glomerular filtration**[47] by presumably compressing the GBM through their adhesion to the GBM and tensile forces of their cytoskeleton, which in turn reduces the permeability to macromolecules.[48,49] Furthermore, they regulate glomerular filtration through the formation of the slit diaphragm and by sensing the glomerular filtration pressure by a mechanoreceptor complex situated at the slit diaphragm.[50]
5. **Clearance of the glomerular filtration barrier** podocyte foot processes exhibits a high abundance of clathrin-coated pits and multivesicular bodies,[51] showing high endocytic activity.[52] Many proteins required for slit diaphragm integrity are recycled through endocytosis, especially nephrin and podocin.[53] Glomerular filtration barrier clearance from immunoglobulins occurs through neonatal Fc receptor-mediated endocytosis[54] and from albumin by transcytosis.[55]

Current Concept of Podocyte Function and Glomerular Filtration

Glomerular filtration is ensured by the concerted interplay of GEnCs, MCs, podocytes, and the GBM on the one hand and the driving filtration pressure modulated by the respective vascular resistances of the afferent and efferent glomerular arterioles on the other hand. Recently, an electrokinetic model of filtration has been proposed, whereby filtration pressure establishes a streaming potential across the glomerular filtration barrier with the Bowman space being more negatively charged than the endothelial lumen. This might establish a retrograde electrophoretic field that acts opposite to diffusive and convective fluxes and tends to exclude negatively charged macromolecules away from the glomerular filtration barrier during active filtration.[107]

Combining quantitative morphologic analyses with mathematical modeling, podocytes are now thought to compress the GBM to counteract filtration pressure and thus prevent leakage of albumin.[49] As such, alteration of buttress force of podocytes (i.e., in the setting of angiotensin 2 signaling) results in decreased compression of the GBM and albuminuria.[49]

PODOCYTE PATHOPHYSIOLOGY

Due to their localization, molecular, and metabolic setup, podocytes are permanently targeted by, and required to respond to, various physiologic and pathophysiologic stressors. If exposure is too excessive in time and degree, this leads to complex adaptive and maladaptive intracellular changes, leading to the typical histopathologic sequence of foot process effacement, podocyte hypertrophy, and podocyte detachment from the GBM with loss into the urine (Fig. 4.3). Podocyte dysfunction results in clinical proteinuria and in a variety of glomerular responses, such as disruption of podocyte-endothelial crosstalk and activation of podocyte-parietal cell interactions culminating in glomerulosclerosis.

More than 80 pathways that result in podocyte distress in individuals with inherently unique and divergent genetic backgrounds have been described. They include circulating factors, cell-surface signaling, mechanical (circumferential and shear) stress, metabolism, proteostasis, fibrosis, inflammation, and the actin cytoskeleton.[56,57] As a general theme, podocyte injury appears to involve reactivation of developmental programs such as those engaged by Notch,[58] Wnt,[59-62] mTOR,[63] and Hippo[64] pathways. Overactivation, imbalance, and impairment of these central intracellular signaling pathways disrupt normal podocyte cytoskeletal regulation, energy metabolism,[65] and protein homeostasis,[66] thus initiating a mostly irreversible dedifferentiation process.

Podocyte Foot Process Effacement

Podocyte function largely depends on its complex three-dimensional cytoskeletal structure. The identification of by now more than 80 monogenic causes of podocyte disease[67,68] has immensely increased our understanding of podocyte function and dysfunction. Regardless of the underlying disease, a characteristic and almost predictable and early response to podocyte injury is a change in shape, called effacement.[69] Numerous studies have shown that effacement is an active process due to changes in the actin cytoskeleton of the podocyte, which forms the "backbone" of these highly specialized cells.[70] A functional imbalance among key regulators of the actin cytoskeleton, such as the Rho family of small GTPases including RhoA, CDC42, and RAC1, is usually observed.[57] Further evidence that effacement is an active process is that in some instances it can be reversed, such as in treatment-responsive patients with minimal change disease (MCD). There has been debate as to whether effacement per se causes proteinuria because proteinuria due to podocyte damage can occur independent of this change in shape. The relationship between podocyte foot process effacement and proteinuria is not fully understood yet[71] but might be explained by the direct action of podocytes on the GBM.[49] It is generally accepted that effacement is a manifestation of serious podocyte injury and that this histologic finding implies changes in either slit diaphragm proteins (i.e., nephrin[72] and podocin[73]), actin binding and regulating proteins (i.e., α-actinin-4[74] and CD2AP[75]), podocyte attachment to the

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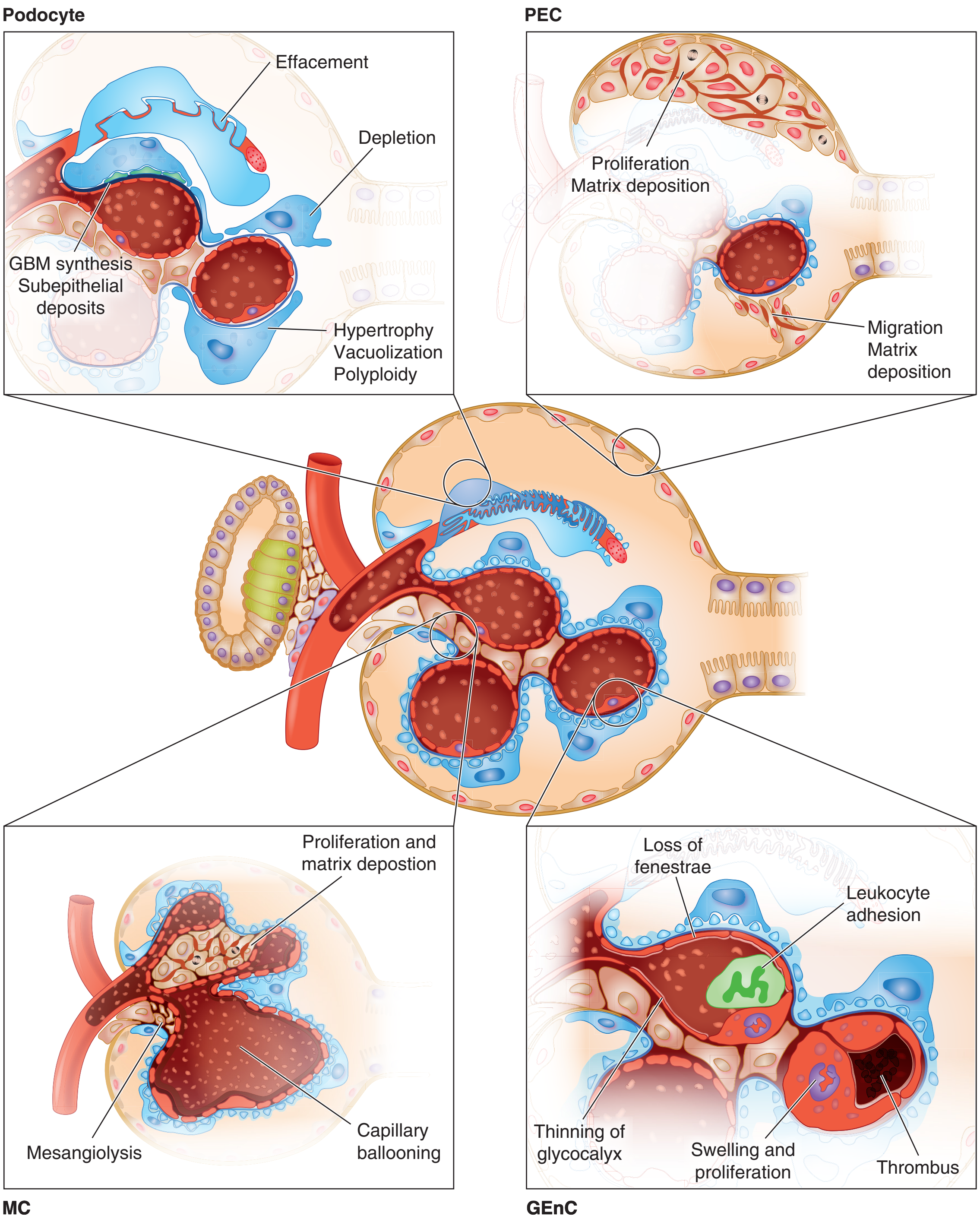

Fig. 4.3 Pathophysiological reaction patterns of glomerular cells. *GEnC,* glomerular endothelial cell; *MC,* mesangial cell; *PEC,* parietal epithelial cell.

GBM (i.e., laminin β2[76] and integrin β4[77]), nuclear proteins (WT1[78] and LMX1B[79]), mitochondrial[80] and lysosomal[81] components, and/or other events, as genetic studies in humans suggest.

Podocyte Hypertrophy

Podocytes are terminally differentiated epithelial cells and unable to adequately proliferate to cover denuded areas of the GBM in situations of glomerular (GBM) distention (i.e., in case of renal hyperfiltration) or podocyte loss. Despite virtual absence of podocyte mitosis and regeneration, current knowledge suggests that differentiated podocytes do have at least some (limited) capacity to adjust to an altered glomerular architecture by hypertrophy. Thereby hypertrophy can be adaptive in the setting of glomerular development, growth, and numerically limited podocyte depletion (up to 20%) or reflect a multifactorial maladaptive response of podocytes due to persistent injury-promoting stimuli such as high glucose in diabetic nephropathy or subepithelial deposits in membranous nephropathy (MN).[82] Recent findings highlight mammalian target of rapamycin (mTOR) and its downstream target the translational

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repressor protein 4E-BP1[83] as key regulators of both adaptive and maladaptive podocyte hypertrophy, whereby the timing, extent, and duration of mTOR activation decides whether hypertrophy is adaptive or maladaptive.[63] Inhibition of mTOR by rapamycin in the setting of adaptive hypertrophy results in proteinuria and glomerulosclerosis, whereas inhibition of mTOR in the setting of maladaptive hypertrophy could be of therapeutic benefit.[84] Besides an imbalance of mTOR signaling pathways,[85,86] podocyte hypertrophy in response to hyperglycemia and stretch has been shown to be also mediated by the cyclin-dependent kinase inhibitor p27Kip1.[87,88] Hypertrophy in MN seems to originate in part from altered protein degradation and subsequent cytoplasmic accumulation of proteins.[89] Hypertrophic podocytes may also be unable to maintain a normal foot process structure,[90] increasing local shear stress, which triggers podocyte detachment.[91]

Podocyte Depletion

Many podocyte diseases are accompanied by a progressive decline in overall kidney function, measured clinically by a decrease in glomerular filtration rate. This is largely due to glomerulosclerosis, with or without tubulointerstitial fibrosis. Patterns of glomerulosclerosis histologically include a segmental form (a portion of an individual glomerulus is scarred) and the more extensive global form (most of an individual glomerulus scars). Podocyte depletion is a major contributor to the development of age-related glomerulosclerosis in humans and rodents.[92-95] A decrease in podocyte number is one of the best predictors of a poor outcome in clinical diabetic kidney disease. A loss of up to 20% of podocytes is tolerated by rats[96] and mice[97] and is accompanied by mesangial cell proliferation and expansion. Segmental glomerulosclerosis ensues when 40% of podocytes are depleted and global glomerulosclerosis when podocyte number is below 60% of normal.[96] There has been a long-standing debate on the underlying mechanisms for podocyte depletion, ranging from necrosis, apoptosis, and necroptosis to the detachment of viable cells from the GBM.[98] Interestingly, viable podocytes can be isolated out of the urine of proteinuric patients, emphasizing the importance of podocyte detachment as a mechanism of podocyte depletion in glomerular diseases.

Studies have suggested that despite a lack of proliferation, podocyte number can be restored following certain therapies such as angiotensin-converting enzyme inhibition.[99] Despite compelling data, it remains unclear whether and to what extent podocyte regeneration exists in renal aging or in pathophysiologic situations and, if podocyte regeneration exists, from what source of resident progenitor cells the novel podocytes originate.[100] Possible sources discussed might be progenitor cells derived from glomerular PECs[101] and/or cells of renin lineage,[102] although further studies are needed to fully validate these findings.

Podocyte-Related Mechanisms of Proteinuria

Proteinuria is the clinical hallmark of podocyte injury. Any kind of injury affecting podocyte function results in proteinuria, either selective as albuminuria (mostly loss of the 60 kDa protein albumin) or nonselective as global proteinuria (loss of a multitude of proteins over 60 kDa of size, including immunoglobulins of 150 kDa) in case of major breakdown of the glomerular filtration barrier. Genetic studies and animal studies have demonstrated that podocyte-dependent proteinuria originates from:

1. <u>Alteration of the slit diaphragm</u> through hereditary or acquired defects of one or more structural slit diaphragm proteins, such as nephrin or podocin, leads to increased passage of proteins across this barrier. Either an absolute decrease in slit diaphragm protein levels or a change in their subcellular location is associated with proteinuria. Moreover, given the complex interplay between proteins comprising the slit diaphragm, a change in one slit diaphragm protein often leads to a cascading dysfunction of one or more of the other proteins.[103]
2. <u>Alteration of the podocyte cytoskeletal network (genetic or acquired)</u> culminating in enhanced or decreased actin polymerization, ultimately leading to proteinuria through a loss of the complex three-dimensional structure and flexibility of the podocyte cytoskeleton.
3. Although no known hereditary mutations of podocalyxin have been described, loss of podocalyxin results in <u>reduced negative surface charge</u> of podocyte processes and proteinuria.[30]
4. <u>Podocyte depletion</u> (i.e., through loss of adhesion)[104] results in denuded GBM areas through which proteins escape.
5. <u>Alteration in GBM composition</u> by podocytes through increased secretion of extracellular matrix proteins is typically observed in membranous and diabetic nephropathy. These extracellular matrix proteins are laid down along the GBM and eventually lead to the characteristic thickening of the GBM in these diseases.[105] The altered extracellular matrix composition leads to secondary changes (loss) in negative charge to the GBM, thereby enabling increased passage of proteins. In addition, increased production and secretion of reactive oxygen species and metalloproteinases from podocytes leads to degradation of the GBM and proteinuria.
6. <u>Effect on the glomerular endothelial cell,</u> the survival of which depends in part on the vascular endothelial growth factor A (VEGF-A) produced and secreted from podocytes.[106] A decrease in production by podocytes, such as when podocyte number decreases, leads to secondary apoptosis of GEnCs, which in turn is accompanied by a decrease in resistance of this layer of the glomerular filtration barrier.

MESANGIAL CELLS

STRUCTURE

Mesangial cells (MCs) are traditionally divided into intraglomerular and extraglomerular MCs, which together comprise four MC subpopulations with distinct transcriptome profiles.[108] MCs are mainly derived from the metanephric mesenchyme[109] and migrate into the cleft of the comma and S-shaped bodies, as well as the maturing glomerulus under the chemotactic control of platelet-derived growth factor B[110] and the survival factor VEGF, both secreted from the progenitors of podocytes and GEnCs.[111] In the mature glomerulus, MCs constitute the central stalk and are in continuity with the extraglomerular mesangium and the juxtaglomerular

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apparatus. Extraglomerular MCs are in close connection to afferent and efferent arteriolar cells by gap junctions allowing for intercellular communication.[112] MCs are highly branched with processes extending in all directions. First, MCs have major processes that contain abundant bundles of microfilaments, microtubules, and intermediate filaments.[113] These major processes contain actin, myosin, and α-actinin and connect MCs with anchoring filaments to the GBM opposite podocyte foot processes and at paramesangial angles, giving them contractile properties.[114] Second, MCs have abundant microvillus-like projections arising from the cell body or the major processes. Within the cell body microfilament bundles are less frequent and the perinuclear region is free of microfilaments.[113] MCs are in direct contact with GEnC without an intervening basement membrane on the capillary lumen side, where the cell membranes of both cell types interdigitate.[115] MCs express genes that classify MCs as a special form of microvascular pericytes,[116] as well as of fibroblasts and vascular smooth muscle cells, suggesting that they represent a hybrid of pericyte-vascular smooth muscle cells and fibroblasts.[108] Single-cell RNA sequencing approaches identify MC marker genes,[117] such as *Pdgfra* and *Itga8* in humans and mice.[108] MCs are positive for vimentin and desmin in mice and COL6A1 in humans.[108]

FUNCTION

Mesangial cells together with their matrix form a functional unit with GEnCs and podocytes.[8]

1. MCs are required for the development and **for the mechanical (structural) support of glomerular capillary loops.**[118,119] Mechanical support is in part mediated by attachment of MCs to the carboxy-terminus globular domain of laminin α5 in the GBM.[120]
2. MCs **regulate the glomerular capillary flow and ultrafiltration surface** and hence the fine-tuning of glomerular filtration by cell contraction at the single nephron level.[121] Contraction of MCs is regulated by vasoactive substances[122] and is dependent on the calcium signaling response and membrane permeability.[122,123] Relaxation of MCs, on the other hand, is mediated by paracrine factors, hormones, and cAMP, with a role for growth factor priming.[122]
3. MCs are responsible for the **homeostasis of the mesangial matrix** through synthesis and degradation of their own matrix components (type IV collagen α1 α2, type V collagen, laminin α,β1, β2, fibronectin, proteoglycans [heparan and chondroitin sulfate, decorin, and biglycan], entactin, and nidogen).[110,124-126] The mesangial matrix provides structural support to the glomerulus[8] and regulates the behavior of MCs such as growth and proliferation[127] partly by binding and sequestering growth factors, influencing their activation and release.[8] Furthermore, the mesangial matrix signals to MCs in response to mechanical stretch.[128]
4. MCs are **source and target of growth factors, cytokines, and vasoactive agents.**[8] For example, MCs produce TGF-β, VEGF, and CTGF[8] in response to capillary stretching resulting from glomerular hypertension. MC proliferation, eicosanoid, and matrix production are influenced by PDGF-B,[129] PDGF-C,[130] fibroblast growth factor (FGF),[124] hepatocyte growth factor (HGF),[125] connective growth factor (CTGF),[131] epidermal growth factor (EFG),[132] and TGF-β.[126]
5. MCs keep the mesangial space free from accumulating macromolecules, which trespass from the capillary lumen through the fenestrated endothelium. For this purpose, MCs are equipped with an elaborate lysosomal system[36,133] and **phagocytose glomerular basal lamina or immune complexes formed at or delivered to the glomerular capillaries.**[134,135] They aid neutrophils to phagocytose apoptotic cells.[136] Macromolecules are removed by both receptor-dependent and receptor-independent mechanisms, depending on the size, charge, concentration, and affinity for MC receptors.[8,137]
6. MCs are **involved in the tubuloglomerular feedback** by communicating with vascular smooth muscle cells over gap junctions.[138]
7. MCs **maintain endothelial health and function** by cross communication, via the mediators and pathways described earlier.

PATHOPHYSIOLOGY

To date, no discrete primary disease of mesangial cells has been described. However, MCs react to changes in the intravascular milieu (soluble factors), immunoglobulin deposition, and changes affecting GEnCs and podocytes. Glomerular mesangial cell injury is commonly associated with mesangial immune deposit formation in IgA nephropathy, lupus nephritis, and Henoch-Schönlein purpura (also designated as IgA vasculitis).[139] Even though IgA nephropathy primarily affects the mesangium (mesangioproliferative disease), it produces marked hematuria and proteinuria, indicating changes in permeability of endothelial cells, GBM, and podocytes. The myriad of biologic responses of mesangial cells to injury range from mesangiolysis to mesangial hypercellularity, mesangial expansion, and the promotion of glomerular inflammation (Fig. 4.3).

Mesangiolysis is defined as a dissolution or attenuation of mesangial matrix and degeneration of mesangial cells by either apoptosis or lysis without obvious damage to the capillary basement membranes. The matrix swells, loosens, and eventually dissolves; the MCs may show edema and vacuolization. Mesangiolysis results in a dilation of the glomerular capillary lumina, as the mechanic support of capillaries provided by the anchoring points of MCs to the GBM is lost.[140] Loss of intraglomerular MCs can be replenished by in-growth of extraglomerular MCs.[141]

Mesangial hypercellularity is characterized by an increase of intraglomerular mesangial cell number by hypertrophy, proliferation, and migration[109] as in IgA nephropathy. In cases where the proliferative insult is limited, mesangial hypercellularity is limited by apoptosis and phagocytosis of these apoptotic cells by adjacent MCs or infiltrating inflammatory cells.[142]

Mesangial expansion is a term characterizing the widening of the intraglomerular mesangium. Mesangial expansion occurs in diabetic nephropathy by excess mesangial matrix production such as fibronectin by MCs[143] or decreased degradation of mesangial matrix by metalloproteinases.[144] Mesangial expansion also occurs through the deposition of immune complexes, light

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chains, amyloid, fibrils, complement, and worn-out GBM material.[145]

Mesangial mechanosensitive injury signaling occurs in response to glomerular hyperfiltration, encompassing the activation of mechanosensitive transcriptional regulators, including coactivators myocardin-related transcription factor A and B (MRTFA/B), yes-associated protein 1 (YAP1 or YAP), and the transcription factor serum response factor (SRF).[146] The MRTF-SRF mechanosensitive pathway affects smooth muscle proliferation, actin cytoskeleton, integrin cell-surface interactions, extracellular matrix organization, and organ fibrosis.[147,148]

Promotion of Glomerular Inflammation

Injured MCs generate reactive oxygen species, proinflammatory activators such as platelet-activating factor (PAF),[149] cytokines (TNF-α, CSF-1, and IL-6), and chemokines,[8] thus sustaining and perpetuating glomerular inflammation.

GLOMERULAR ENDOTHELIAL CELLS

STRUCTURE

Glomerular capillaries are highly permeable to water and small solutes while maintaining relative impermeability to macromolecules, potentially even as small as albumin.[150] GEnCs are highly specialized cells that form the continuous inner layer of glomerular capillaries. GEnC have a particular embryonic origin, with the majority of GEnCs arising by vasculogenesis from mesenchymal precursors in combination with a minority of GEnC arising from introgression of existing vessels.[151,152] In the mature glomerulus, the nucleus of GEnC bulges into the capillary lumen. The cytoplasm of GEnC is 200 nm thin at its slimmest areas and punctuated by numerous fenestrae. The fenestrae of GEnC are the largest in comparison with the fenestrae of endothelial cells of other organs and represent circular pores of 60 to 80 nm in diameter, which cover 20% to 50% of the glomerular endothelial surface.[153,154] GEnC fenestrae are hourglass in shape with the smallest diameter of the fenestrations being midway between the apical and basal surfaces[155]; hence the actual area of GBM available for filtration at the base of fenestrae is greater than that of the smallest diameter.[155] Under special fixation conditions, a diaphragm that spans the fenestrae is visible.[156] In quiescent GEnCs, only 2% of total glomerular capillary cross-section has diaphragmed fenestrae.[157] The functional significance is the allowance of bidirectional transfer of small or soluble substrates between blood and the extracellular space,[158] as well as a high fluid flux driven by hydrostatic pressure through the fenestrations with negligible resistance to water permeability.[153] The universal component of fenestral diaphragms is plasmalemmal vesicle-associated protein 1 (PLVAP), which is low in mature quiescent GEnCs compared with GEnCs recovering from injury.[157] Fenestrae maturation depends on ADAM10-Notch signaling[159] and VEGF-A.[106] GEnCs are covered by a 200- to 400-nm thick glycocalyx, which represents a negatively charged gel-like surface structure of proteoglycans with their covalently bound polysaccharide chains named glycosaminoglycans (GAGs), glycoproteins, and glycolipids. The main carbohydrate constituents are heparan sulfate (HS), chondroitin sulfate (CS), and hyaluronan (HA) bound to the hyaluronan binding surface proteins such as CD44. The glycocalyx is attached to the GEnC by charge-charge interactions,[160] rendering GEnC sensitive to hemodynamic factors[150] such as shear stress.[161] The glycocalyx covers the fenestrae and the interfenestral domains of GEnC equally[162]; however, the thickness of the glycocalyx differs between fenestrated and nonfenestrated GEnCs.[163] The intrafenestral glycocalyx has a higher concentration of heparan sulfate, which is important for permeability properties (i.e., to restrict albumin passage).[164] The GEnC fenestrae are plugged by a high content of hyaluronan[163] and are considered to be key components of the glomerular permeability barrier. Like other endothelial cells, GEnC expresses the specific markers platelet endothelial cell adhesion molecule 1 (PECAM1, CD31), intercellular adhesion molecule 2 (ICAM2), vascular endothelial growth factor receptor 2 (VEGFR2), and growth factor receptor Tie2, von Willebrand factor (vWF), and vascular endothelial (VE)-cadherin (CD144).[165] Single-cell RNA sequencing analyses classify GEnCs by the expression of *Ehd3, Kdr, Igfbp5, Emcn,* and *Tmem204.*[108] GEnC homeostasis strongly depends on a specialized proteasome constitution initially identified in immune cells, the immunoproteasome system.[36,166]

FUNCTION

1. GEnCs are involved in the **GBM production** together with podocytes but to a lesser extent than podocytes, as seen in mice with podocyte-specific type IV collagen α5 chain deletion (Alport model), which exhibit marked thinning and alteration of the GBM.[167,168]
2. GEnCs contribute to the **hydraulic conductivity** of the glomerular filtration barrier through GEnC fenestrations,[169] which are formed in response to VEGF-A[165,170] in downstream Rac and ERK1/2 dependent signaling pathways.[164]
3. GEnCs contribute to the **size and charge selectivity of the glomerular filtration barrier** by the endothelial surface lining composed of the membrane-bound glycocalyx and the loosely bound endothelial cell coat.[163,171] The glycocalyx adds to size and charge selectivity most likely by forming a meshlike structure of negatively charged HS and the less charged hyaluronan,[172] as infusion of enzymes that degrade the glycocalyx increases albumin passage through the GEnC.[173]
4. The **glycocalyx of GEnC protects against protein leakage, inflammation, and coagulation.**[173] With properties like a hydrogel, the glycocalyx acts as a size barrier to protein. With the high negative charge of polyanions, the glycocalyx electrically repels proteins.[174-176] GAG degrading enzymes such as chondroitinase and heparanase alter glomerular permeability.[177,178] The gel-like antiadhesive properties of the glycocalyx preclude the interaction of leukocytes with adhesion molecules.

PATHOPHYSIOLOGY

GEnCs are primarily targeted or involved in several forms of kidney thrombotic microangiopathies, including vasculitis, hemolytic uremic syndrome, and preeclampsia.[179] Even though GEnCs are primarily affected, these conditions are associated with mesangiolysis and proteinuria, indicating endothelium-dependent changes in MCs and podocytes.

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Injury of the GEnC induces the release of vasoactive substances, changes in the composition of the endothelial glycocalyx and of endothelial adhesion molecules resulting in a net prothrombotic state (Fig. 4.3).[180,181] Furthermore, GEnCs hypertrophy, proliferate, go into apoptosis, and detach, further accelerating thrombosis of glomerular capillaries.[180] Experimental models indicate that recovery from glomerular injury is dependent on GEnC angiogenesis.[182-184] Visualization of GEnC injury remains challenging. Morphologic signs of GEnC injury are swelling of the cell body, thinning of the glycocalyx,[185,186] loss of fenestrations, and enhanced expression of adhesion markers such as CD34[187] and E-selectin (CD62E), an adhesion receptor for leukocytes. GEnC injury has been demonstrated to arise from an altered crosstalk with podocytes.

Endotheliosis

Glomerular capillary endotheliosis describes the swelling of GEnC with the deposition of fibrous material in and beneath GEnC, a condition typically seen in preeclamptic glomerular injury. The net result of GEnC swelling and deposition of fibrous material is capillary occlusion.

Changes of Glycocalyx and Their Sequelae

Since the glycocalyx functions as a molecular scaffold and binds 1. circulating proteins such as growth factors and chemokines; 2. proteins involved in cell attachment, migration, and differentiation; and 3. proteins involved in blood coagulation and inflammation, alterations of the glycocalyx are central to the pathology of GEnC and glomerular injury. GEnC injury induces changes in both the thickness and molecular composition of the endothelial glycocalyx. Changes in glycocalyx thickness are mainly due to upregulation of glycocalyx-degrading enzymes such as hyaluronidase, heparanase, and proteinases, thereby shedding glycocalyx fragments into the glomerular circulation. Alteration of glycocalyx composition such as decreased levels of hyaluronan or changes in HS sulfation patterns change the antiadhesive and anticoagulative properties of the glycocalyx. In total, loss of glycocalyx thickness and glycocalyx modification result in enhanced permeability of the filtration barrier to proteins, inflammation, and coagulation.[173]

Mediation of the Inflammatory Reaction

Injured GEnC release vasoactive substances (nitric oxide and endothelin), which regulate glomerular filtration. Furthermore, GEnCs promote glomerular inflammation by attracting leukocytes by means of shed glycocalyx fragments and by expressing leukocyte adhesion molecules. Upon perturbation of the glycocalyx, these adhesion molecules become unmasked and allow leukocyte interactions with the endothelial surface. HS of the glycocalyx acts as a direct ligand for L-selectin.[188] Upon stimulation with inflammatory stimuli, GEnCs increase the expression of HS domains,[189] which facilitates leukocyte extravasation.[190] The endothelium binds chemokines, which regulate the extravasation of leukocytes upon a chemoattractant gradient.[191] Alterations of glycocalyx composition ensue in enhanced binding of chemokines by way of positive charge interactions.[192]

PARIETAL EPITHELIAL CELLS

Parietal epithelial cells (PECs) are derived from the metanephric mesenchyme and line the BC of the renal glomerulus. During the vesicle and comma stages in glomerular development, PECs share a common phenotype with the other epithelial cells of the later glomerulus, namely podocytes and proximal tubular cells. With formation of the Bowman space in the S-shaped body, the phenotypes of PECs, podocytes, and proximal tubular cells diverge. In the mature kidney, PECs are a heterogeneous population of cells with five distinct PEC populations discerned by single-cell RNA sequencing.[193] PECs at the urinary pole maintain features of proximal tubular cells, and PECs at the vascular pole maintain features of podocytes and are therefore termed parietal podocytes.[194]

STRUCTURE

PECs resemble squamous epithelial cells with their small thin cell bodies ranging in thickness from 0.1 to 0.3 μm increasing to 2.0 to 3.5 μm at the nucleus. The surface of some PECs is lined by microvilli and cilia in a range from 0 to 2 cilia per cell.[195] They are interconnected by "labyrinth-like" delicate tight junctions located at the apical surface, which comprise claudin-1, claudin-2, claudin-3, K-cadherin (Cdh6), and kidney-specific cadherin (Cdh16), occludin, and zonula-occludens 1 (ZO-1).[196] PECs have a simple cytoskeleton with filaments at the basal membrane region.[197] They express the intermediate filament protein cytokeratin 8.[198] PECs have the transcriptional prerequisite to express podocyte markers.[199] The expression of podocyte proteins is negatively regulated through protein degradation[199] and by microRNA-193a,[200] which represses WT1 mRNA levels.[201] PECs express the transcription factor Pax2 from the paired box family, which is involved in regulating genes governing proliferation, cell growth, and survival. Furthermore, PECs can be differentiated from podocytes through the expression of EPH receptor A7 belonging to the ephrin receptor subfamily, ladinin (a proposed anchoring filament that is a component of basement membranes), scinderin (a calcium-dependent protein that regulates cortical actin networks),[202] and the glycoprotein Dickkopf 3 (DKK3) and lipopolysaccharide-binding protein (LBP).[108]

FUNCTION

1. PECs at the vascular pole constitute a **potential reservoir for podocytes** in glomerular development, maturation,[97,203] and eventually even adulthood.[204]
2. PECs presumably **form the basement membrane of the Bowman capsule**, which consists of laminin-111, laminin-511,[205,206] type IV collagen α1 α2, and type IV collagen α5 α6,[207,208] but this is not definitely proven.
3. PECs **prevent the leakage of urine from the urinary space into the periglomerular compartment.**[209]

PATHOPHYSIOLOGY

To date, there is no evidence for glomerular injuries related to primary PEC injury. However, PECs have taken the center stage of attention for their contribution to glomerular diseases such as rapid progressive glomerulonephritis (RPGN) and focal segmental glomerulosclerosis (FSGS).

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Activated PECs exhibit a larger cytoplasm and larger rounder nuclei. Further signs of PEC activation are a de novo expression of CD44[210] and the phosphorylation of signaling molecules.[211,212] The predominant reactions of PECs to glomerular injury are proliferation, migration, and synthesis of matrix that results in the development of crescents or glomerular tuft scars (Fig. 4.3). There is an ongoing debate whether PECs serve as an intrinsic fixed progenitor population to replenish podocytes or proximal tubular progenitor cells in glomerular regeneration.[213] PECs respond to injury with the expression of podocyte markers such as synaptopodin and WT1.[214,215]

Proliferation and Migration

Activation of PECs as in RPGN results in their proliferation and the formation of extracapillary proliferations, also called cellular glomerular crescents, of which PECs are the major constituents.[216] Crescent formation is the result of not only PEC proliferation but also partly the transdifferentiation of PECs to myofibroblasts,[210,217] the deposition of matrix, and the infiltration with inflammatory cells. Crescents can occlude the tubular outlet of the glomerulus, resulting in entire nephron degeneration,[218] and are generally associated with poor prognosis. PEC proliferation is usually associated with glomerular endothelial cell, GBM, or podocyte injury, as leakage of plasma components such as fibrin from the blood circulation is a strong inducer of PEC proliferation.[219,220] Additionally, PEC depletion initiates PEC proliferation and crescent formation.[221] Migration of PECs from the BC onto the glomerular tuft is the predominant reaction of PECs in FSGS.[216] In this scenario, PECs are activated and invade the glomerulus at focal areas via adhesions connecting sclerotic capillary areas with the BC.[222] PECs are then present in the sclerotic regions[223,224] and can be visualized by staining for CD44.[225] Potentially therapeutic amenable pathogenic upstream signals of PEC activation are CD9,[226] macrophage migration inhibitory factor (Mif), and colony-stimulating factor 1 receptor (Csf1).[193]

Matrix Deposition

Thickening of the BC due to matrix production can be observed in aging glomeruli[227] and glomerular injury. Matrix deposition by PECs is observed in glomerular crescents and in the sclerotic glomerular tuft regions in FSGS. PECs that migrate to the glomerular tuft produce and deposit extracellular matrix proteins. The composition of this extracellular matrix is related to that of the BC[223] and contains specific HS moieties of HS proteoglycans.[228]

GLOMERULAR CELL CROSSTALK

Podocytes, glomerular endothelial cells, mesangial cells, and PECs with their respective matrix must be considered as a functional unit, in which every cell plays its part in ensuring proper glomerular filtration. Consequently, 30 years of isolated cell-type–based research is now being replaced by systems biology approaches, to integrate the role and contribution of each individual glomerular cell type for the proper glomerular biology and function. Recent advances have demonstrated that glomerular cell crosstalk is the prerequisite for normal glomerular development and health. Furthermore, primary injury of one glomerular cell type or of the GBM affects the other glomerular cell types by crosstalk. Clinical and experimental observations suggest that crosstalk exists between podocytes and glomerular endothelial cells, between glomerular endothelial cells and mesangial cells, between mesangial cells and podocytes, and finally between podocytes PECs (Fig. 4.4). Crosstalk occurs 1. through the secretion of ligand(s) by one glomerular cell type that binds to their cognate receptor on the other glomerular cell type and 2. by the secretion of extracellular vesicles (EVs, which contain proteins, lipids, mRNA, microRNA, DNA) by one glomerular cell type, which are taken up by and thus modify the other glomerular cell type.[229]

PODOCYTES—GLOMERULAR ENDOTHELIAL CELLS

Podocytes are essential for GEnC development and maintenance, and the first glomerular cell crosstalk to be identified was the cross communication from podocytes to GEnC. Podocytes secrete vascular growth factors, such as VEGF-A, which binds to its cognate receptor VEGFR2 on GEnC, a crosstalk crucial for GEnC health[106] and which, if disrupted, results in glomerular injury. Tight control of VEGF-A levels is essential for correct glomerular barrier function, and even though other podocyte-specific proteins such as the transcription factor Pod1/Tcf21[230] or transforming growth factor (TGF)-β activated kinase 1 (Tak1)[231] have been shown to be essential for GEnC development and health, it remains to be established whether this is a direct effect or a consequence of altered VEGF-A levels. Podocyte progenitors also express ephrin B2, another vascular growth factor, and this might contribute to EphB4 receptor expressing GEnC development and health.[232] Angiopoietin-1 (Angpt1) is expressed by both podocytes and MCs and binds to the tyrosine-protein kinase receptor (Tie2/Tek) expressed on GEnC. This crosstalk is thought to stabilize the glomerular capillaries, as mice with induced deletion of Angpt1 at embryonic day 10.5 exhibit dilated capillary loops and disrupted subendothelial GBM structures and reduced MCs, whereas podocytes appeared intact.[233] GEnC homeostasis is further maintained by podocyte expression of the glucocorticoid receptor, loss of which modulates GEnC expression of α-smooth muscle actin, TGFβR1, and β-catenin.[234] Podocytes also secrete the chemokine CXCL12 (SDF1), which binds to its receptor CXCR4 on GEnC, a crosstalk important for the formation of glomerular capillaries.[44]

Despite their importance in glomerular development and maintenance, some podocyte-derived signals have been shown to perpetuate or attenuate GEnC injury in pathologic settings. Enhanced circulation of endothelin-1 or enhanced podocyte expression of endothelin-1, which binds in a paracrine fashion to the endothelin receptor A on GEnC, mediates mitochondrial oxidative stress and dysfunction in adjacent GEnC.[235,236] Furthermore, endothelin 1 induces podocytes to release the GEnC glycocalyx-degrading enzyme heparanase, contributing to GEnC injury in diabetic nephropathy.[237] Another example for disease-perpetuating crosstalk is the podocyte-specific expression of angiopoietin-2, which results in GEnC apoptosis without affecting podocytes.[238] The CXCL12/CXCR4 crosstalk enhances GEnC injury in diabetic nephropathy[239] and in shiga toxin–associated hemolytic uremic syndrome.[240] A

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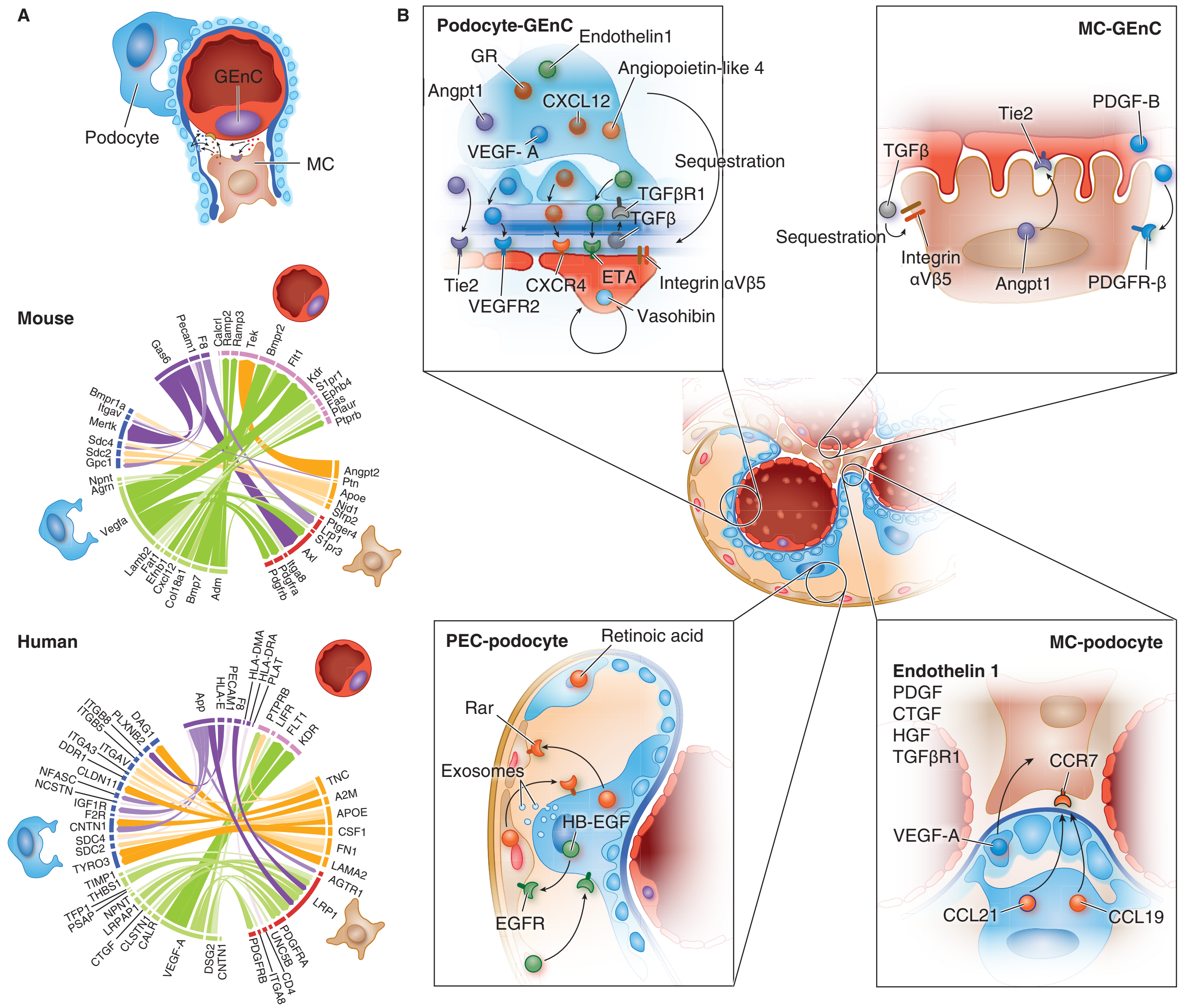

Fig. 4.4 Intraglomerular crosstalk. (A) Chord diagrams showing ligand/receptor interactions among MCs, GEnCs, and podocytes identified in mouse and human scRNA-seq datasets. Colors correspond to cell types and ligand/receptor categories: MCs (ligands: orange, receptors: dark red), GEnCs (ligand: purple, receptors: hot pink), and podocytes (ligands: lime, receptors: blue). The thickness and opacity of the arcs are proportional to the weights of ligand-receptor interactions as used in the NicheNet model. (B) Examples of intraglomerular crosstalk. Angpt1, angiopoietin 1; CTGF, connective tissue growth factor; CXCL12, C-X-C chemokine ligand 12; CXCR4, chemokine receptor 4; EGFR, epidermal growth factor receptor; Eta, endothelin-1 receptor A; GR, glucocorticoid receptor; HB-EGF, heparin-binding epidermal growth factor–like growth factor; HGF, hepatocyte growth factor; PDGF, platelet-derived growth factor; PDGF-B, platelet-derived growth factor B; PDGFRβ, platelet-derived growth factor receptor beta; Rar, retinoic acid receptor; TGFβ, transforming growth factor beta; TGFβR1, transforming growth factor beta receptor 1; Tie2, tyrosine-protein kinase receptor 2; VEGF-A, vascular endothelial growth factor A; VEGFR2, vascular endothelial growth factor receptor 2. (Modified from He B, et al. *Nat Commun*. 2021;12:2141.)

GEnC protective crosstalk with podocytes was suggested for podocyte-derived angiopoeitin-like-4 (Angptl4), which is structurally similar to angiopoietins but does not signal over Tie2. Angptl4 was suggested to protect GEnC from oxidative injury in nephrotic syndrome by binding to α5 β5 integrins.[42] A protective reverse signaling from GEnC to podocytes has been demonstrated for vasohibin secreted from GEnC, which is thought to counteract VEGF-A signaling in situations of pathologic elevated VEGF-A levels such as in diabetic nephropathy.[241]

GLOMERULAR ENDOTHELIAL CELLS—MESANGIAL CELLS

The fate of MCs and GEnC is tightly linked. Both communicate directly in the paramesangial areas of glomerular capillaries, where their plasma membranes are in direct contact. Even though MCs secrete a multitude of factors in vitro, which could affect GEnC, few MC-secreted factors have been identified that participate in GEnC crosstalk in vivo. This is in part a consequence of the lack of glomerular endothelial cell and of mesangial

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cell-specific gene targeting strategies in vivo, which are prerequisites for such investigations. Endothelial-derived PDGF-B, which binds to its MC-expressed receptor PDGFR-β, has been demonstrated to be of crucial importance for the development and maintenance of glomerular capillaries.[8] Consequently, injury and loss of GEnCs (e.g., due to toxin- or antibody-related GEnC injury) results in decreased PDGF-B levels and MC death (mesangiolysis). MCs maintain endothelial health and function through integrin αV β8-dependent sequestering of TGF-β, thereby reducing the amount of active TGF-β.[242] Furthermore, like podocytes, MCs synthesize angiopoietin-1, which binds to its receptor Tie2 on GEnC[233] to stabilize the vasculature. Ligand-receptor interaction studies from single-cell RNAseq data suggest crosstalk of MCs with GEnCs via LDL receptor-related protein 1 (LRP1) as a multifunctional scavenger involved in phagocytosis with F8 (coagulation factor 8) and amyloid precursor protein (APP).[108]

PODOCYTES—MESANGIAL CELLS

It is not clear yet at which sites podocytes communicate with MCs. In vivo evidence of podocyte–mesangial cell crosstalk is scarce. Nonetheless, experimental data and clinical observations in several hereditary forms of nephrotic syndrome due to mutations in podocyte-specific genes[243] suggest such a communication exists. For example, in glomerular development, mutations in podocyte genes such as the transcription factor Pod1/tcf21,[244] phospholipase Cε1[245] laminin α5[120] and of Wilms tumor antigen[78] result in a failure of MCs to migrate into glomeruli. The podocyte-specific deletion of collagen type IV α3 (Alport mouse) results in enhanced expression of integrin α1 by MCs,[246] possibly affecting MC cell adhesion and cell signaling at the GBM. The chemokines CCL19 and CCL21, generated by podocytes, bind to CCR7 on MCs and are thought to regulate local MC migration and adherence to the GBM.[247] Also, a decrease of VEGF-A secretion from podocytes results in mesangiolysis,[45] supporting the idea of a podocyte-mesangial cell crosstalk, which is of relevance in glomerular development. In glomerular injury, experimental data support a communication of podocytes with MCs, although it cannot be excluded that the observed effects on MCs in podocyte injury are not the result of altered PDGF-B levels related to podocyte-dependent GEnC injury. Several signaling pathways might be involved in podocyte-MC crosstalk in the setting of injury, such as endothelin 1, PDGF, CTGF, HGF, and TGF-β.[43]

PODOCYTES—PARIETAL EPITHELIAL CELLS

Under physiologic conditions, podocytes and PECs are in close proximity at the vascular pole, where transitional cells called parietal podocytes carry both the characteristics of PECs and podocytes.[194] Another theoretical site of cross communication is across the Bowman space, where the apical membranes of podocytes and PECs can overcome the physical separation given by the primary filtrate and touch. Controlled depletion of PECs results in transient proteinuria with focal podocyte foot process effacement,[221] and podocytopenia is associated with PEC hyperplasia,[248] suggesting interdependence of both cells. Communication between PECs and podocytes could also ensue from the uptake of podocyte-derived proteins from the primary urine by PECs.[249] Furthermore, podocytes release exosomes to the urine,[250,251] which in turn could affect PECs.

In glomerular injury, podocytes are in close contact to PECs by bridges formed between the capillary tuft and the BC[252] and in intraglomerular crescents, of which they are both constituents.[216,253] In glomerulonephritis, mathematical three-dimensional multiscale modeling studies[254] and experimental data suggest that podocyte and PEC cross communication might regulate proliferation of both cells and regeneration of podocytes from PECs. Proliferation of both cell types and thereby crescent formation was shown to be dependent on the heparin-binding epidermal growth factor–like growth factor (HB-EGF), which is de novo expressed by podocytes and PECs in rapid progressive glomerulonephritis and the EGF receptor, which is found on both cells.[255] Lineage tracing experiments suggest that podocyte regeneration in glomerulonephritis occurs from renal progenitor cells located in the BC[101] and that this process can be enhanced by retinoic acid.[256] Thereby, retinoic acid synthesized in glomeruli promotes renal progenitor cells to differentiate toward a podocyte phenotype.[257]

COMMON MECHANISMS OF GLOMERULAR DISEASES

There appear to be several basic responses of the glomerulus to injury such as cellular proliferation, changes in glomerular cell phenotypes, and increased deposition of extracellular matrix. Any cause of severe glomerulonephritis (GN) can cause crescent formation (typical for rapid progressive GN), which is composed of parietal cells,[221,258] podocytes,[253] and inflammatory cells.[259,260] Most forms of glomerular injury result from immunologic mechanisms,[261-265] which include both humoral and cellular components. Glomerulopathies resulting from direct viral cytotoxicity have been demonstrated for HIV-1[266] and are debated for SARS-CoV-2.

The exact causes of the immune responses that lead to GNs are often unknown, but it is believed that autoimmunity, drugs, toxins, and infections may trigger similar pathways, causing GN.

The **humoral response** is often a T-helper cell 2 (Th2)-mediated response resulting in B cell activation, antibody generation and deposition, and complement activation. Immunoglobulin and complement component deposition and found in most human glomerular diseases, suggesting that the humoral response is crucial in the development of glomerular injury, which has been the rationale for the use of therapeutic B cell depletion in various glomerular diseases. There are three patterns of immunoglobulin deposition in the glomerulus: 1.) Immune deposits in the GBM and in the subepithelial space (underneath podocytes) are typical for MN and usually do not initiate a strong inflammatory reaction, as the deposits are separated from the circulation by the GBM; 2) Immune deposits in the subendothelial space (lupus nephritis and membranoproliferative GN); or 3) (IgA nephropathy and lupus nephritis), on the other hand, both initiate multiple inflammatory processes. The final pattern of immunoglobulin deposition is determined by the biologic properties of the immunoglobulins (IgG subtype) deposited, the absolute amount of

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immunoglobulins deposited, and the mechanisms whereby the deposits are formed. Deposition of the complement-fixing IgG1 or IgG3 subtypes usually results in more severe glomerular injury than deposition of IgA or IgG4, which both activate complement poorly.[267] Principally, antibody-antigen binding that takes place within the glomerulus to glomerular self- or non-self-antigens (termed in situ binding) induces an immune response depending on the localization of the antigens. Typical glomerular self-antigens are the phospholipase A2 receptor (PLA_2R1)[15] and thrombospondin type-1 domain-containing 7A (THSD7A)[16] in MN or the noncollagenous domain of the α3 chain of type IV collagen known as the Goodpasture antigen.[268,269] Non-self-antigens localize to glomerular capillaries by mechanisms such as charge affinity for glomerular structures or pure passive trapping in the glomerular sieve in form of the antigen alone (termed as planted antigens) or as antigen-antibody complexes formed outside the kidney. In situ antibody binding to planted non-self-antigens is typical in lupus nephritis to DNA nucleosome complexes[270,271] or in IgA nephropathy to abnormally glycosylated IgA.[272] Glomerular deposition of preformed immune complexes to non-self-antigens has been demonstrated in early childhood MN, where immune complexes containing cationic bovine serum albumin are deposited[273] or in hepatitis C virus–associated membranoproliferative GN, where virus-containing cryoglobulins are deposited.[274] There is little experimental evidence that immunoglobulin binding alone induces significant tissue injury, except for when the antibodies bind to podocyte antigens such as nephrin[275,276] of the slit membrane or to PLA_2R1[277] and THSD7A,[278] which induce noninflammatory podocyte injury. Of note, severe inflammation can occur with only small amounts of antibody deposited, as in antineutrophil cytoplasmic antibody (ANCA)-associated GN.

The **cellular response** is a largely T-helper cell 1 (Th1)–mediated response characterized by the infiltration of circulating mononuclear cells such as lymphocytes and macrophages into glomeruli and the formation of crescents. **Neutrophils** are the earliest cells to be found in inflamed glomeruli in human biopsies and are strong inducers of glomerular injury.[279] Animal studies demonstrate that their strongest attractants to inflamed glomeruli are interleukin 8 and complement factor C5a,[280] bound to the glomerular endothelium via HS proteoglycans.[189,281] Neutrophils are activated by phagocytosis of immune complex aggregates, which induces them to undergo a respiratory burst with generation of reactive oxygen species such as hydrogen peroxide. Hydrogen peroxide interacts with the neutrophil cationic enzyme myeloperoxidase (MPO) to halogenate the glomerular capillary wall.[282] Furthermore, neutrophils store other cationic enzymes such as proteinase 3 (PR3), elastase, and cathepsin G, which upon release further degrade the glomerular capillary wall. Lastly, neutrophils release extracellular traps, weblike DNA structures expulsed from nuclei with adherent histones, proteases, peptides, and enzymes, which show a modest contribution to glomerular injury in anti-GBM glomerulonephritis[283] but could be more injurious in ANCA-associated GN and lupus nephritis.[284]

Macrophages are typically found in glomerular lesions with crescents and serve as effector cells of both humoral and cell-mediated forms of immune glomerular injury because their localization to inflamed glomeruli is induced by interactions with immunoglobulins through their Fc receptors and through chemokines, such as RANTES, and macrophage chemoattractant protein 1 (MCP1). Similar to neutrophils, macrophages generate direct tissue injury by release of oxidants and proteases. Additionally, they release tissue factor to induce glomerular fibrin deposition and crescent formation and transforming growth factor beta (TGF-β) to induce the synthesis of extracellular matrix culminating in glomerular sclerosis.[285]

T cells are rarely to be found in injured glomeruli except for GNs primarily mediated by macrophages such as crescentic GN. T cell–mediated glomerular injury mostly results from released chemokines and recruitment of macrophages.[286] However, ovalbumin-specific $CD4^+$ and $CD8^+$ T cells together can induce glomerular injury in transgenic mice that express the antigen ovalbumin in podocytes.[249] Among the known T cell subtypes, there is strong experimental evidence for the importance of T-helper cell 17 (Th17) in crescentic GN.[287,288] Th17 cells produce and secrete interleukin 17 (IL-17) A, IL-17 F, IL-21, and IL-22, which promote inflammation by directly causing tissue injury and enhancing secretion of proinflammatory cytokines and chemokines by resident cells. This results in augmented infiltration of leukocytes, in particular neutrophils recruited by CXCL5[289] to the affected kidney, where they induce further inflammation and injury.[290] The kidney infiltrating Th17 cells are partly recruited from the gut.[291]

Tissue-resident lymphocytes are ideally positioned to quickly respond to pathogens and other environmental stimuli. The kidney harbors several classes of innate and innate-like lymphocytes (ILC) that have been described to contribute to this tissue-resident population.[292] In the kidneys of humans and mice, IL-33 receptor-positive ILC2s are a major ILC subtype that, after effective renal expansion by IL-33 treatment, are considered to be central regulators of renal repair mechanisms.[293] Other tissue-resident cells, however, such as pathogen-induced tissue-resident memory T(RM)17 cells, aggravate glomerular injury,[294] or in the case of kidney-resident γ/δ T cells limit local *Staphylococcus aureus* growth during chronic infection and provide enhanced protection against reinfection.[295]

Platelets are present in glomerular lesions in which intracapillary thrombosis is involved, typically observed in thrombotic microangiopathies and antiphospholipid syndrome. Platelets are important players in the formation of thrombi and recruitment of leukocytes to the inflamed glomerulus.[296] In addition they release factors, which enhance glomerular permeability to proteins, enhance immune complex deposition,[297-299] and induce mesangial cell proliferation (PDGF)[300] and mesangial cell sclerosis (TGF-β).[301]

Dendritic cells (DCs) are restricted to the tubulointerstitium and are absent from glomeruli.[302] However, proteins that pass the glomerular filter are captured by renal DCs or reach the renal lymph nodes by lymphatic drainage[300] to induce immune tolerance[303] to innocuous proteins such as food antigens or hormones or to stimulate infiltrating T cells to produce proinflammatory cytokines.[249]

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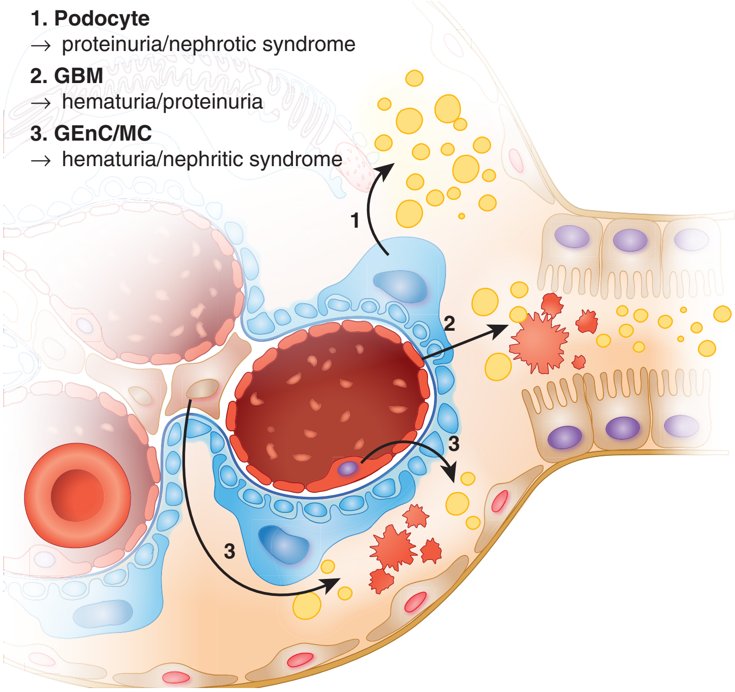

Fig. 4.5 The clinical presentation reflects the localization of a glomerular injury. Injury of podocytes results in proteinuria (yellow droplets) eventually leading to the clinical manifestation of nephrotic syndrome. Injury of the glomerular basement membrane (GBM) commonly results in proteinuria and hematuria with dysmorphic erythrocytes. Injury of glomerular endothelial cells (GEnC) and mesangial cells (MC) usually leads to hematuria with dysmorphic erythrocytes and little proteinuria.

The site of glomerular injury, especially which glomerular cell is involved, determines whether the patient has an inflammatory or a noninflammatory injury (Fig. 4.5). Because glomerular endothelial and mesangial cells are in contact with circulating factors such as complement and inflammatory cells, they are prone to react to injury via a principally more dramatic inflammatory response. In contrast, PECs and podocytes are separated by the GBM from the circulation, so podocyte injury is rarely associated with activation of circulating inflammatory cells. Clinically, the distinction between inflammatory and noninflammatory injury is crucial for the adequate diagnosis and management of patients. The clinical characteristics of **inflammatory injury** are hematuria with dysmorphic erythrocytes with or without red blood cell casts and occasional leukocyturia. Inflammatory injury is accompanied by varying degrees of proteinuria, which ranges from mild to nephrotic range proteinuria and normal or reduced glomerular filtration rate, depending on the severity of disease. Morphologically, inflammatory injury is characterized by glomerular hypercellularity that results from proliferating resident glomerular (mostly mesangial cells and PECs) and from infiltrating hematopoietic cells (mostly neutrophils and macrophages), phenotype change, and visible structural injury. Glomerular injury arises from the release of inflammatory substances from infiltrating hematopoietic cells and from glomerular cells or from the impairment of protective mediators such as complement factor H[304] or complement factor H–related protein 5[305] as negative regulators of the complement pathway. The release of inflammatory substances such as cytokines, growth factors, proteases, products resulting from complement activation (C5a, C5b-9), vasoactive agents, and oxidants[262,306,307] initiates thrombosis, necrosis, and crescent formation, which, if extensive, leads to the serious clinical condition of rapid progressive glomerulonephritis. **Noninflammatory lesions** usually involve podocytes and are termed "podocytopathies." They are characterized by proteinuria and (if proteinuria is extensive) nephrotic syndrome (a triad comprising proteinuria over 3.5 g/day, edema, and hypertriglyceridemia) without hematuria.

MECHANISMS OF INJURY IN COMMON PODOCYTOPATHIES

Common causes of immune-mediated glomerular injury are MCD, primary FSGS, and MN. All three entities exhibit a dramatic increase in glomerular permeability with little to significant structural abnormalities visible by light microscopy. These diseases are classified as podocytopathies, as podocytes are thought to be the primary glomerular cell affected in the pathogenesis. In contrast, "nontraditional" podocyte diseases include diabetic kidney disease, human immunodeficiency virus nephropathy, amyloidosis, Fabry disease, membranoproliferative glomerulonephritis, and postinfectious glomerulonephritis. The inciting causes of each podocyte disease differ, and therefore each disease affects podocytes in different ways; in turn, the response to injury in each disease differs, leading to different histologic and clinical manifestations. In diabetic nephropathy, for example, a glomerulopathy with podocyte involvement,

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dysfunction, and injury occurs at the level of glomerular cell crosstalk, resulting in single nephron hyperfiltration (for further discussion see Chapter 41). Regardless of the inciting causes and their mediators, several common clinical and pathologic responses occur in podocyte injury, as highlighted earlier, namely hypertrophy, foot process effacement, loss, and proteinuria.

MINIMAL CHANGE DISEASE AND FOCAL SEGMENTAL GLOMERULOSCLEROSIS

CHARACTERISTICS OF MCD

In MCD the glomerulus is per definition mostly normal by light microscopy with absence of complement and no or only discrete immunoglobulin deposition. In light of the histologic "minimal changes," pathologic changes are best being evidenced by electron microscopy with foot process effacement, microvillous transformation, and vacuolization. The absence of glomerular sclerosis differentiates MCD from FSGS. MCD typically presents as a steroid-sensitive nephrotic syndrome, contrasting FSGS, which often presents with steroid-resistant nephrotic syndrome. Despite the overlapping characteristics of MCD and FSGS, they might represent a spectrum of diseases rather than a complete merging of two disease entities.

CHARACTERISTICS OF FSGS

Focal segmental glomerulosclerosis (FSGS) is a generic term for a histologic injury pattern defined by segmental glomerular consolidation into a scar that affects some but not all glomeruli with a wide range of etiologic interpretations. FSGS describes both a disease characterized by primary podocyte injury (primary FSGS) and a lesion that occurs secondarily in any type of chronic kidney disease (secondary FSGS).[308] There is abundant evidence that classical FSGS is the consequence of podocyte loss in experimental models,[309] which is accompanied by proliferation and migration of PECs to the glomerular tuft (both discussed earlier). The underlying causes or mechanisms of FSGS are broadly considered as hereditary/congenital and sporadic/acquired in nature. Primary FSGS presents with either steroid-sensitive nephrotic syndrome or steroid-resistant nephrotic syndrome. Secondary FSGS, on the other hand, is associated with nephron loss, drug toxicity, or viral infections and rarely presents with nephrotic syndrome. It is, however, steroid resistant.

PATHOPHYSIOLOGIC CONCEPTS OF MCD AND PRIMARY FSGS

There is significant overlap in the factors causing these podocytopathies, which are the following:

1. **Soluble serum factors:** On the basis of experimental data, soluble serum factors are thought to be causative in MCD and FSGS, as nephrotic plasma has direct cellular effects on cultured podocytes[310] or in single perfused kidneys.[311] Many efforts have been undertaken to unravel "the increasing or missing circulating permeability factor."[312] Potential candidates include TNFα,[313,314] circulating cardiotrophin-like cytokine factor 1 (member of the IL-6 family), circulating hemopexin,[315,316] and the soluble urokinase-type plasminogen activator receptor (suPAR)[317] in the development of nephrotic syndrome.[318] These factors, however, need further verification in the context of human disease activity,[319,320] disease specificity,[321,322,323] and therapeutic effects.[324]
2. **Immune dysfunction:** Considerable clinical and experimental evidence points toward an immune dysfunction on T and B cell sides in MCD. Clinically, MCD is responsive to not only steroids but also rituximab,[325,326] a monoclonal antibody against plasma cell CD20, and a significant association exists with HLA-DQA1 (a MHC class II) missense coding variants.[327] Additionally, MCD is associated with Hodgkin disease and allergies.[328] On the experimental side, rodents develop proteinuria if they receive $CD34^+$ peripheral stem cells from MCD patients[329] or after injection of the supernatant derived from T cells or from peripheral blood mononuclear cells from affected patients.[330] T and B cell dysfunction is further suggested by a different DNA methylation pattern of Th0 cells,[331] altered Th17/regulatory T cell balance,[332] and upregulation of T cell–derived interleukin 33 and 13 is suggested in patients with MCD,[333,334] the latter causing proteinuria and foot process effacement in rats.[335,336] Further pointing toward immune dysfunction as the origin of nephrotic syndrome is the finding that abatacept, an antibody challenging the CD80 (B7-1)-CTLA-4-axis, has been shown to reduce proteinuria in FSGS,[337] an effect still requiring further confirmation.[338-340] Experimental findings have given rise to the idea of disease-specific expression of CD80[337,341] or expression of a hyposialylated form of angiopoietin-like 4 by podocytes, which might contribute to glomerular disease.[342-344]
3. **Anti-nephrin antibodies:** Since the discovery and cloning of nephrin as a key component of the slit diaphragm,[72,345,346] anti-nephrin antibodies have been described in experimental models,[347,348] recurrent FSGS,[276,349] type 1 diabetes,[350] and MCD.[351] A recent multicenter study using novel quantitative detection methods found anti-nephrin autoantibodies in 69% of MCD cases and 90% of idiopathic nephrotic syndrome cases without immunosuppressive treatment, with levels correlating with disease activity.[352] An experimental model demonstrates that anti-nephrin antibodies induce nephrotic syndrome and podocyte dysfunction.[352] Together, these findings suggest that many cases of nephrotic syndrome could be reclassified as antinephrin autoantibody podocytopathy.
4. **Genetic inheritance:** The biologic functions altered by the gene mutations involved in FSGS in podocytes are broad, ranging from cytoskeletal regulation, slit diaphragm function, lysosomal function, mitochondrial function, and attachment to the GBM.[67] In the late 1990s positional cloning of the gene responsible for congenital nephrotic syndrome of the Finnish type led to the identification of the archetypal podocyte-specific protein, nephrin.[72] This was rapidly followed by the identification of other proteinuric diseases linked to podocyte-specific single-gene disorders including those affecting podocin,[73] Wilms tumor 1,[353] CD2AP,[354] α-actinin-4,[74] TRPC6,[355] phospholipase Cε1 (PLCE1),[356] WW

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and PDZ domain-containing 2 (MAGI2),[357] and kidney ankyrin repeat-containing protein (KANK).[358,359] Whole-exome sequencing data from the Toronto glomerulonephritis registry showed that 20% of FSGS patients may have a monogenic etiology of their disease, with *COLA4* disorders being the most prevalent.[68] In each of these conditions it is generally accepted that proteinuria results directly from the disruption of these constitutively expressed genes in the podocyte, leading to FSGS. Even though the amount of by now more than 60 known mutations of podocyte genes is steadily increasing, these mutations are rare and explain less than 30% of patients with hereditary cases and only 10% to 20% of patients with sporadic cases of FSGS.[67] Studies of the increased susceptibility of African Americans to FSGS have implicated variants of another gene expressed in podocytes, apolipoprotein L1 (APOL1) as a genetic modifier.[360,361] Experimental expression of the APOL1 risk alleles (termed *G1* and *G2*) in a podocyte-specific manner demonstrated that these were causal for podocyte foot process effacement, proteinuria, and glomerulosclerosis.[362] Mechanistically, the risk-variant APOL1 alleles interfere with endosomal trafficking and block autophagic flux, ultimately leading to inflammatory-mediated podocyte death and glomerular scarring.[362] For more discussion on genetic renal diseases, see Chapter 44.

5. **Altered posttranslational regulation of podocyte proteins:** Besides abnormalities in podocyte genes, altered posttranslational regulation of podocyte mRNA by small noncoding RNA molecules, termed "microRNAs," have been shown to result in FSGS, such as the transcription factor Wilms tumor protein 1 (WT1) by microRNA-193a[201] or of Notch1 and p53 by microRNA-30.[363] The involvement of other miRNAs in podocyte (patho)physiology is becoming more and more appreciated, but a direct causative effect to FSGS remains to be shown. As such, microRNA145-5-p is thought to reach podocytes via exosomes and to affect podocyte apoptosis.[364,365]

MEMBRANOUS NEPHROPATHY

Membranous nephropathy (MN) is an autoimmune disease with the morphologic hallmarks of GBM thickening, granular staining for human IgG, and complement components along the glomerular filtration barrier and subepithelial (subpodocyte) electron-dense deposits by electron microscopy. While this histologic pattern can arise from deposition of preformed immune complexes as in lupus nephritis type 5 or from subepithelial antigen trapping in hepatitis B virus and hepatitis C virus–associated secondary forms of MN, primary MN is thought to be the consequence of in situ binding of autoantibodies to podocyte-expressed antigens. The concept that primary MN is an antibody-mediated disease has been supported by the discoveries of autoantibodies to podocyte membrane antigens such as neutral endopeptidase (NEP),[366] the phospholipase A_2 receptor (PLA_2R1),[15] and thrombospondin type 1 domain-containing 7A (THSD7A).[16] PLA_2R1 and THSD7A are targets for a malfunctioning immune system in 70% and 5% of adult cases, respectively, and NEP is important in a small number of neonates with MN caused by alloimmunization due to vertical transfer of antibodies from a genetically NEP-deficient mother.[267] The causes of autoimmunity are likely to be multiple as mirrored by the multitude of recently discovered (potential) MN autoantigens such as neural epidermal growth factor–like 1 (NELL1), or exostosin 1/exostosin 2,[367-369] and based on the clinical correlative observations of genetic predispositions,[370] tumor-associations,[371] and association with bacterial infections,[372] among others.[367,373] PLA_2R1 polymorphisms influence the susceptibility to MN, and the association between certain HLA-DQA1 alleles and MN suggests that these HLA class II molecules could facilitate autoimmunity against PLA_2R1.[370] In contrast to PLA_2R1-associated MN, THSD7A-associated MN has a high cooccurrence of malignancy.[371,374] The direct pathogenicity of autoantibodies is suggested by observations that PLA_2R1 or THSD7A autoantibodies are present in patients with rapid recurrence of MN in renal transplants,[16,375,376] the finding that anti-PLA_2R1 antibody levels are associated with disease remission[377,378] and progression,[378-380] and the finding that MN can be induced in mice, which normally express THSD7A on podocytes[381,382] by injection of human autoantibodies[278] or rabbit antibodies to THSD7A.[383] Proof that PLA_2R1-specific autoantibodies are pathogenic was for a long time hampered by the fact that rodents necessary for such studies normally do not express PLA_2R1 on podocytes. However, injection of PLA_2R1-containing patient serum to minipigs enabled proof of pathogenicity.[277] The role of complement in the pathogenesis of human MN remains unclear, even though components from all three complement pathways (alternative, classical, and lectin) are found in renal biopsies from patients with MN.[384] First, the deposited IgG in idiopathic MN is typically of the noncomplement-fixing IgG4 subtype. Second, in rodent models of MN, clinical and morphologic MN can be induced in the absence of detectable complement deposition,[278,383] as well as in rodents with genetic deficiency in complement components.[385,386] On the other hand, C3 and the membrane attack complex C5b9 are usually constituents of the deposits.[387] C5b9 inserts into the podocyte membrane and is transported across the cell[388] and excreted into the urine, where high levels of C5b9 can be measured in humans.[389] Experimental studies allow for an injurious and protective function of C5b9. Sublytic C5b9 has been shown to induce podocyte injury by multiple pathways such as the activation of kinases, the induction of endoplasmic reticulum stress, and the production of extracellular matrix.[390] On the other hand, C5b9 enhances the ubiquitin-proteasome system,[391] which supports a protective removal of damaged proteins. Recent investigations in experimental autoimmune THSD7A MN demonstrate a primary complement activation via the classical pathway, a process leading to exacerbation of nephrotic syndrome.[384]

EFFECTS OF EXISTING THERAPIES ON PODOCYTES

The basis for therapy of primary nephrotic syndrome usually includes antihypertensive and antiproteinuric therapies and dietary recommendations.[392] Immunosuppressive treatments vary on the basis of diagnosis, and therapies specifically targeting podocytes are under investigation. For more detail on therapies for specific forms of nephrotic syndrome, see Chapter 30.

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RENIN-ANGIOTENSIN SYSTEM BLOCKADE

Blockade of the renin-angiotensin system belongs to the standard supportive therapy regimen for primary podocytopathies with proteinuria. In addition to altering glomerular hemodynamics, the systemic and glomerular upregulation of the renin-angiotensin system (RAS), primarily via angiotensin II, plays a role in the development of proteinuric diseases. Elevated tissue levels of angiotensin II and its angiotensin type 1 receptor are noted in glomeruli[393] and podocytes[394] in primary podocytopathies. Activation of RAS is detrimental to glomerular cells including podocytes as it promotes multiple trophic effects such as extracellular matrix protein accumulation, reactive oxygen species production, oxidative stress, alteration in slit diaphragm proteins partly by epigenetic modulation of nephrin promoter methylation,[395] or by caveolin-1 mediated downregulation and dephosphorylation of nephrin,[396] Rho/ROCK-associated disruption of the podocyte cytoskeleton,[397] increased calcium influx through TRPC6 channels,[398,399] or calcium release from intracellular stores such as store-operated calcium channels (SOCs),[400] cell cycle inhibition, detachment and inflammatory cytokine production,[394,401] and p38-MAPK activation of podocyte apoptosis.[402] Blocking the RAS with angiotensin-converting enzyme inhibitors, angiotensin 1 receptor (AT1R) antagonists, and mineralocorticoid receptor blockers reduce proteinuria resulting in renoprotection, an effect that is attributed to a reduction in glomerular hydrostatic pressure and an abolishment of the detrimental trophic glomerular effects mentioned earlier.[47]

SODIUM GLUCOSE TRANSPORTER TYPE 2 INHIBITORS (SGLT2i)

Initially developed as an antidiabetic treatment strategy, blockade of the proximal tubular glucose reabsorption by the SGLT2 has evolved into a new therapeutic approach to glomerular/podocyte injuries due to the profound antiproteinuric effects of SGLT2i and their significant cardiovascular and renal protection[403,404] going beyond their modest antihypertensive action.[405-407] SGLT2i are effective in long-term kidney health by altering glomerular hemodynamics through tubulo-glomerular feedback modulation, ultimately decreasing single nephron hyperfiltration,[408] and by this glomerular mechanical stress. Besides these effects, SGLT2i might exert direct podocyte protection, as it reduces podocyte cytoskeletal rearrangement, proteinuria, and podocyte loss.[409] Low-level SGLT2 expression by podocytes is discussed later.[409-411] In the diabetic setting, SGLT2i ameliorates mTORC1-associated podocyte injury,[412] glomerular inflammation, matrix expansion, and podocyte loss,[412,413] and a reduction of podocyte lipotoxicity is observed.[410,413] Additionally, SGLT2i modifies glomerular crosstalk, especially pathogenic paracrine VEGF-A signaling from podocytes, which induces GEnC injury.[414]

ENDOTHELIN-1/ENDOTHELIN A RECEPTOR ANTAGONISTS

In glomeruli and podocytes, the vasoactive peptide endothelin-1 exerts many deleterious effects, the predominant one being glomerular hypertension through renal vascular vasoconstriction, which is mediated by ET-A and ET-B receptors.[415] Endothelin-1 further induces nephrin excretion, podocyte foot process effacement, and podocyte loss[416,417] and induces activation of NF-kb and β-catenin signaling.[410] As one of the crosstalk systems between podocytes and GEnCs, the endothelin-1/ET-A system results in GEnC injury.[235,418,419] As ET-A blockade additionally yields antiinflammatory and antifibrotic effects,[417,420] it is an interesting target for therapeutic intervention. Combining the synergistic effects of ET-A and angiotensin II type 1 receptor blockade, Sparsentan, as a dual antagonist exhibits significant antiproteinuric effects.[419]

GLUCOCORTICOIDS

Steroids are immunomodulatory drugs widely used in the treatment of proteinuric diseases, but their modes of action especially in the noninflammatory forms of nephrotic syndrome, such as MCD and primary FSGS, remain completely unknown. Glucocorticoid receptor (GR) expression is ubiquitous; therefore these drugs could affect any glomerular cell type. However, deletion of the GR in GEnCs or podocytes upregulates Wnt signaling and profibrotic gene expression and suppresses fatty acid oxidation in podocytes,[234,421] suggesting that glucocorticoids modulate important homeostatic and injury pathways within the normal podocyte-GEnC crosstalk. It has been shown that GR-induced signaling pathways are functional in murine podocytes, having transcriptional and posttranscriptional effects on podocyte genes.[422] Initial reports in murine[423] and human[424] podocytes showed that dexamethasone exerts potent biologic effects directly on podocyte structure and function. These include limiting podocyte apoptosis[425] and induction of podocyte differentiation by restoring the actin cytoskeleton,[423] increasing levels of the transcription factors Kruppel-like factor 15 (KLF15),[426] and preventing the downregulation of protective microRNA-30.[363] Specifically the slit diaphragm protein nephrin is affected by steroids, as steroids enhance the transport of nephrin from the endoplasmic reticulum[427] and induce the phosphorylation of nephrin,[428] which is reduced in MCD[429] and important for the function of nephrin and thereby of the slit diaphragm. Glucocorticoids also reduce caveolin-1 upregulation and the associated podocyte injury by deletion of epithelial membrane protein 2 (EMP2), of which mutations are a recognized monogenic cause of podocytopathies.[430]

EVIDENCE FOR DIRECT ACTIONS OF IMMUNOSUPPRESSANTS ON PODOCYTES

The **calcineurin inhibitors** cyclosporine and tacrolimus are widely used in nephrotic syndrome, either alone or in combination with other therapies. Calcineurin is a Ca^{2+}-dependent phosphatase, which dephosphorylates nuclear factor of activated T cells (NFAT), a transcription factor. Dephosphorylation of NFAT initiates its cytoplasmic to nuclear translocation, resulting in an increased transcription of genes such as *TRCP6*,[431] whose gain of function mutations in podocytes induce FSGS.[355] In accordance, podocyte-expressed NFAT is a strong inducer of glomerulosclerosis in

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mice.[432] Besides their known immunomodulatory effects in T cells, calcineurin inhibitors affect the podocyte cytoskeleton by transcriptional downregulation of the aforementioned calcium channel TRPC6[433] and by preventing the degradation of the actin-organizing protein synaptopodin[434] in an NFAT-independent manner. Together, these mechanisms appear to result in a stabilization of the podocyte actin cytoskeleton and direct reduction of proteinuria.

The specific **anti-B cell monoclonal antibody rituximab**, increasingly considered to be effective in proteinuric diseases even when they are not all obviously immune mediated,[435] has been shown to have direct effects on podocytes including stabilizing their actin cytoskeleton.[436] Although monoclonal antibodies are assumed to have specific binding targets, they can also have "off-target" effects. In this case it seems that rituximab, as well as binding to the CD20 molecule, which is its accepted molecular target, also binds to a podocyte protein called sphingomyelin phosphodiesterase acid–like 3b (SMPDL-3b), and this protein stabilizes the actin cytoskeleton.[436]

IDENTIFICATION OF CANDIDATE THERAPEUTIC APPROACHES FOR THE FUTURE

End-stage kidney disease constitutes an enormous burden of morbidity, both to patients who suffer a lifelong chronic disease and to health services that are challenged by high costs arising from dialysis and transplantation. Despite major advances over the past 15 years in unraveling genetic and biological causes of podocyte dysfunction as the origin of most forms of nephrotic syndrome, glomerular target cell–directed therapies are still in their infancy. Especially in comparison with other specialties in medicine, only a few new drugs have been approved for renal failure in comparison with many new drugs for cancer therapy or heart disease. Nonetheless, the separation of monogenic causes of podocyte dysfunction from other causes is now used to stratify those patients in whom immunosuppression can be minimized due to low potential for success. Furthermore, the discovery of the genes involved in hereditary proteinuric disease unraveled critical (and in the future potentially druggable) pathways required for podocyte maintenance, stabilization of the podocyte actin cytoskeleton and slit diaphragm, and for restoration of metabolic and mitochondrial function.[437] One such example could be targeting APOL1 with the use of antisense nucleotides[438] or by inhibiting APOL1 G1/G2 protein transmembrane cation channel function.[439,440] Promising results were achieved with the selective APOL1 channel blocker inaxaplin, which reduced proteinuria in participants with two *APOL1* variants and FSGS in a phase 2 study.[441] Several experimental approaches in recent years have aimed at finding podocyte-specific interventions, which stabilize the **actin cytoskeletal backbone** of podocyte foot processes and thereby also the slit diaphragm.[442,443] Targeting the **metabolic and mitochondrial function** in podocytes seems to be another therapeutic option for many forms of podocyte injury. Even though mTOR inhibition is known to induce proteinuria as a typical side effect,[444,445] inhibition of an overactivated mTOR pathway in the setting of metabolic oversupply as in diabetic nephropathy is beneficial for podocyte function and proteinuria.[84] Other examples of targeting metabolic pathways with therapeutic potential include lipid-lowering (e.g., statins),[446,447] antidiabetic agents such as thiazolidinediones (TZDs), or glitazones, which activate the peroxisome proliferator-activated receptor γ (PPARγ)[448,449,450,451] and manipulation of VEGF levels[452] or autophagy.[453] Because podocytes are now recognized as both a source and a target of **complement**-mediated injury with particular relevance to glomerular disease progression, complement-directed therapy options are likely to expand in upcoming years. Besides the selection of specific podocyte injury targets, which can be targeted pharmaceutically or genetically by CRISPR/Cas-9 gene editing (i.e., *COL4A3* and *COL4A5* in Alport Syndrome[454]), therapeutic options will be enhanced by achieving **precise drug delivery** to podocytes. The application of cell-penetrating peptides,[455] overcoming the physical location of podocytes external to the GBM through biomimetic[456,457] or nanoparticle-based drug delivery systems,[458] or through nanobody-mediated targeting of, for example, adeno–associated viruses, represent only a few of the exciting new approaches under development.

SUMMARY

Kidney diseases with glomerular involvement account for the vast majority of end-stage kidney diseases. In the past 2 decades, much has been learned about the glomerulus, a functional and integrated syncytium of four types of glomerular cells, which together ascertain glomerular filtration. Podocytes and glomerular endothelial cells comprise the glomerular filtration barrier. Together, both cells allow for size- and charge-selective glomerular filtration due to their specialized three-dimensional structure, extensive glycocalyx coating, and synthesis of the unique glomerular basement membrane, which is compressed by the buttress forces of podocytes. Mesangial cells regulate glomerular filtration by means of contraction and release of vasoactive substances and maintain the health of glomerular endothelial cells. As highly phagocytic cells, they clear the mesangium from depositing (macro)molecules. PECs build the BC to prevent leakage of the primary urine to the tubulointerstitium, can contribute to glomerular scarring, and are thought to constitute a potential reservoir for podocytes in development, maturation, and eventually adulthood. A cell type–specific view of glomerular physiology and pathophysiology has enhanced our understanding of glomerular cell biology immensely in recent decades. However, the intricate interactions of glomerular cell types are getting more and more center-stage attention as clinical and experimental observations demonstrate that normal functioning of glomerular filtration requires coordinated interaction of all four cell types and that injury of one glomerular cell type usually affects the others. Several clinical and experimental challenges and opportunities lie ahead. Identifying, designing, and delivering glomerular cell-specific therapeutic agents is actively being pursued to enhance efficacy and reduce systemic side effects. Noninvasive diagnostic testing is being keenly studied, such as measuring glomerular cell products in the urine and markers in the serum and urine, which will hopefully

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translate into clinical practice. The past 2 decades have witnessed phenomenal advances in understanding glomerular cell biology in health and disease, which will hopefully soon translate into better therapeutic options for progressive glomerular and renal kidney disease.

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QUESTIONS

PHYSIOLOGY OF FILTRATION

1. Which glomerular structure is not involved in regulating glomerular filtration?
   a. Glomerular endothelial cells
   b. Parietal epithelial cells
   c. Podocytes
   d. Mesangial cells
   e. Glomerular basement membrane

Answer: b

Rationale: Glomerular filtration is regulated by the coordinate interaction of glomerular endothelial cells, podocytes, and mesangial cells. Podocytes and glomerular endothelial cells comprise the glomerular filtration barrier. Together, both cells allow for size- and charge-selective glomerular filtration due to their specialized three-dimensional structure, extensive glycocalyx coating, and synthesis of the unique glomerular basement membrane. Mesangial cells regulate glomerular filtration by means of contraction and release of vasoactive substances and maintain health of glomerular endothelial cells. Parietal epithelial cells built the BC to prevent leakage of the primary urine to the tubulointerstitium and are not primarily involved in glomerular filtration. They can, however, contribute to glomerular scarring and are thought to constitute a potential reservoir for podocytes in development, maturation and eventually in adulthood.

LOCATION INJURY AND CLINICAL PRESENTATION

2. A 26-year-old female patient presents with swollen legs for the past week. She has been using ibuprofen 800 mg 3× a day for 1 month for a shoulder injury. Her examination is significant for blood pressure 130/90 mm Hg and 3+ lower extremity edema. Urinanalyses: exhibits 4+ protein, no blood, 24-hour urine protein: 10.8 gm, no hematuria, serum creatinine 0.9 mg/dL (eGFR >60 cc/min). What is the most likely glomerular location causing this clinical picture?
   a. Mesangial cells (i.e., IgA nephritis)
   b. Glomerular basement membrane (i.e., Alport syndrome)
   c. Glomerular endothelial cell (i.e., eclampsia)
   d. Parietal epithelial cell (i.e., rapid progressive glomerulonephritis)
   e. Podocyte (i.e., minimal change disease)

Answer: e

Rationale: The predominant symptom of massive proteinuria without hematuria in the setting of preserved renal function is characteristic of primary podocyte injury such as in minimal change disease. In this case, minimal change disease might be a result of the use of nonsteroidal antiinflammatory drugs (NSAIDs). Mesangial cell, glomerular basement membrane, and glomerular endothelial cell involvement usually presents with mild to strong hematuria with characteristic dysmorphic erythrocytes. Parietal epithelial cell involvement is reflected by formation of glomerular scars.

MEMBRANOUS NEPHROPATHY

3. Which of the following are podocyte antigens in adult membranous nephropathy?
   a. suPAR
   b. Neutral endopeptidase (NEP)
   c. Phospholipase A2 receptor (PLA_2R1)
   d. Nephrin
   e. Thrombospondin type-1 domain-containing 7A (THSD7A)

Answer: c and d

Rationale: NEP, PLA2R, and THSD7A are all podocyte antigens associated with membranous nephropathy; however, NEP is only found in rare cases of antenatal membranous nephropathy caused by alloimmunization due to vertical transfer of antibodies from a genetically NEP-deficient mother. Autoantibodies to PLA_2R1 and THSD7A are typically found in adult membranous nephropathy. PLA_2R1 is highly specific for membranous nephropathy and found in 70% of patients with membranous nephropathy and associates with disease activity. Autoantibodies to THSD7A are found in 5% of patients with MN and strongly associate with the tumors.

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